

NKPC HSA Bayesian State-Space Estimation: Inverse markup gap

June 23, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100 \log(N_t^{level}) - \overline{100 \log(N^{level})}}{10}.$$

Thus N_t^{model} , $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t - \theta \hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ , θ , θ_0 , and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.639 [0.538, 0.742]	0.639 [0.536, 0.742]	0.615 [0.504, 0.720]	0.602 [0.485, 0.718]
kappa	0.007 [-0.217, 0.228]	NaN	0.012 [-0.217, 0.240]	NaN
kappa_0	NaN	0.010 [-0.217, 0.241]	NaN	0.036 [-0.210, 0.283]
delta	NaN	-0.003 [-0.040, 0.034]	NaN	-0.002 [-0.040, 0.036]
theta	NaN	NaN	-0.022 [-0.359, 0.577]	NaN
theta_0	NaN	NaN	NaN	0.056 [-0.380, 0.472]
gamma	NaN	NaN	NaN	-0.006 [-0.046, 0.032]
rho_1	NaN	0.756 [0.389, 1.067]	0.902 [0.622, 1.118]	0.962 [0.725, 1.195]
rho_2	NaN	-0.022 [-0.313, 0.235]	0.013 [-0.190, 0.221]	-0.037 [-0.261, 0.184]
phi_1	0.664 [0.530, 0.800]	0.664 [0.536, 0.796]	0.663 [0.532, 0.798]	0.664 [0.529, 0.797]
lambda_ez	-0.019 [-0.343, 0.301]	-0.026 [-0.344, 0.288]	NaN	-0.036 [-0.344, 0.273]
n	NaN	-0.022 [-0.083, 0.040]	-0.034 [-0.083, 0.020]	-0.021 [-0.080, 0.041]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 inv_markup: prior=baseline, period=full

Condition. data=inv_markup, prior=baseline, period=full, constraint=unrestricted (unrestricted).

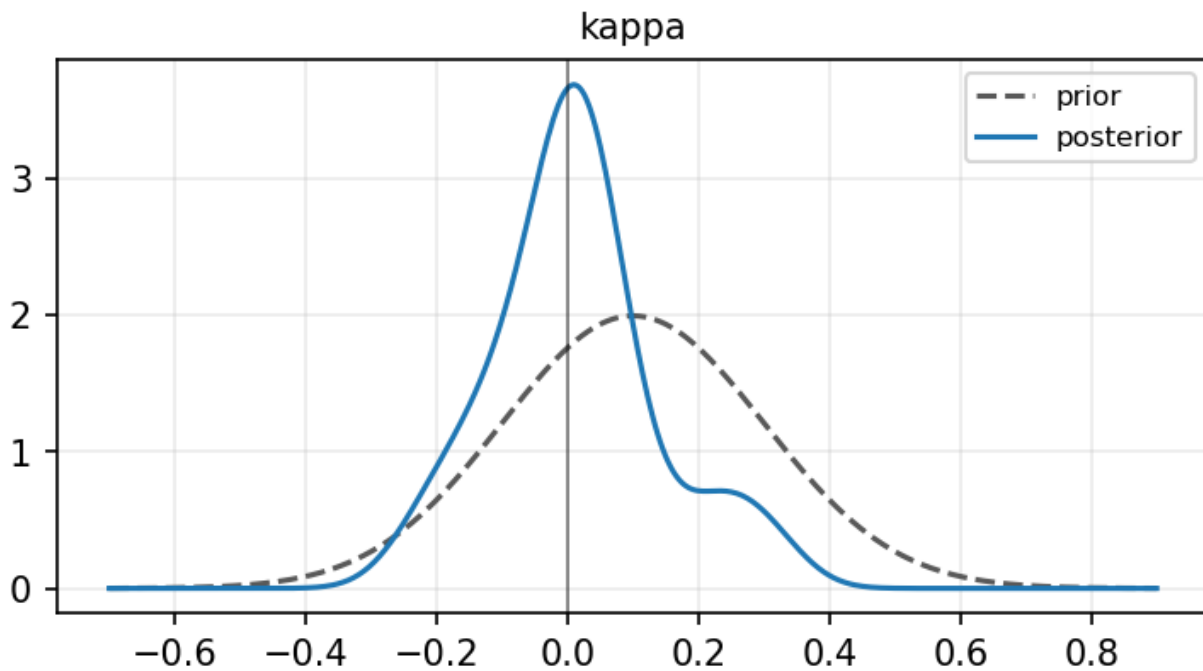
6.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.634 [0.547, 0.743]	0.648 [0.554, 0.776]	0.616 [0.529, 0.724]	0.617 [0.526, 0.720]
kappa	0.012 [-0.205, 0.302]	NaN	0.030 [-0.213, 0.302]	NaN
kappa_0	NaN	0.101 [-0.188, 0.462]	NaN	0.096 [-0.209, 0.405]
delta	NaN	-0.001 [-0.003, 0.002]	NaN	-0.000 [-0.003, 0.001]
theta	NaN	NaN	0.008 [0.003, 0.016]	NaN
theta_0	NaN	NaN	NaN	0.067 [-0.242, 0.371]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.002]
rho_1	NaN	1.805 [1.706, 1.895]	1.871 [1.807, 1.910]	1.191 [1.002, 1.388]
rho_2	NaN	-0.809 [-0.897, -0.709]	-0.875 [-0.915, -0.810]	-0.269 [-0.470, -0.066]
phi_1	0.673 [0.532, 0.774]	0.667 [0.561, 0.778]	0.651 [0.517, 0.783]	0.664 [0.584, 0.784]
lambda_ez	-0.037 [-0.338, 0.225]	0.020 [-0.184, 0.326]	NaN	-0.012 [-0.301, 0.278]
n	NaN	-0.054 [-0.190, 0.050]	-0.035 [-0.114, 0.051]	-0.090 [-0.181, 0.037]

6.2 Prior vs Posterior by Model

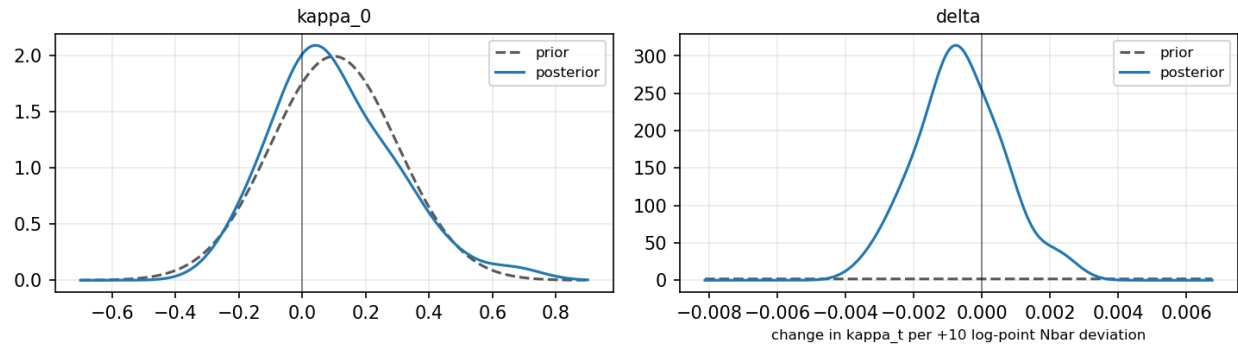
ces

Prior vs posterior — ces



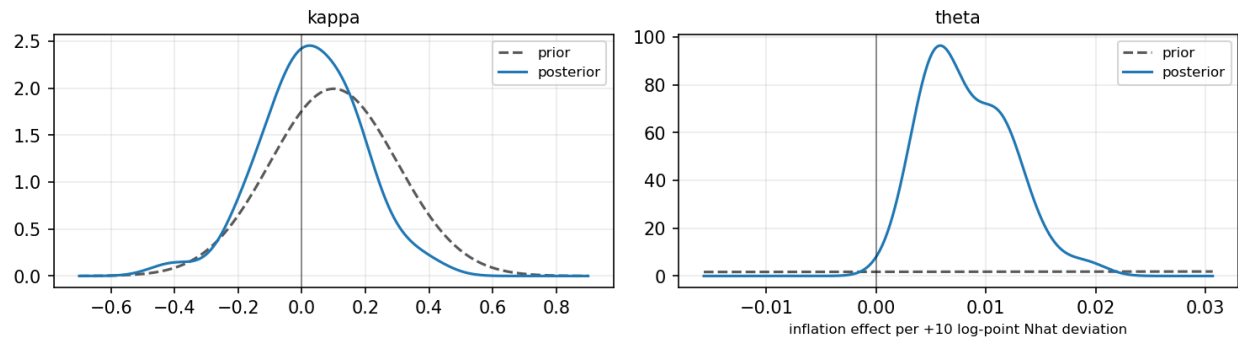
hsa_steady

Prior vs posterior — hsa_steady



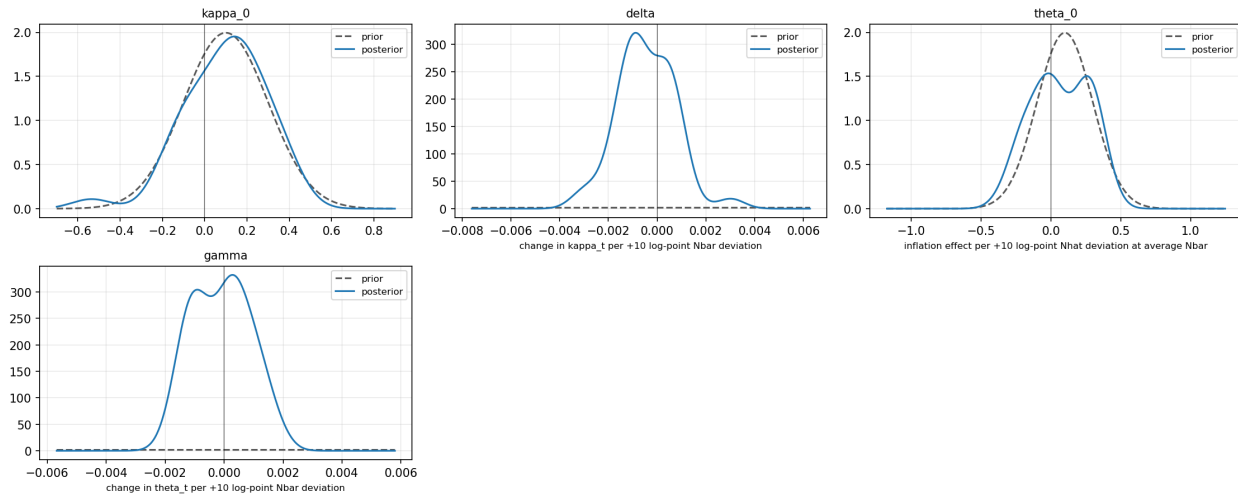
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

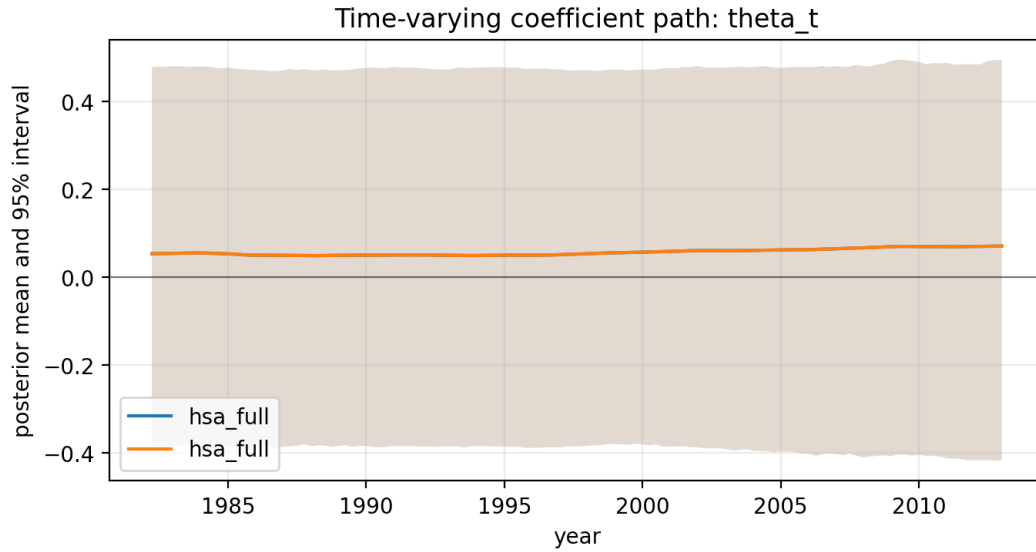
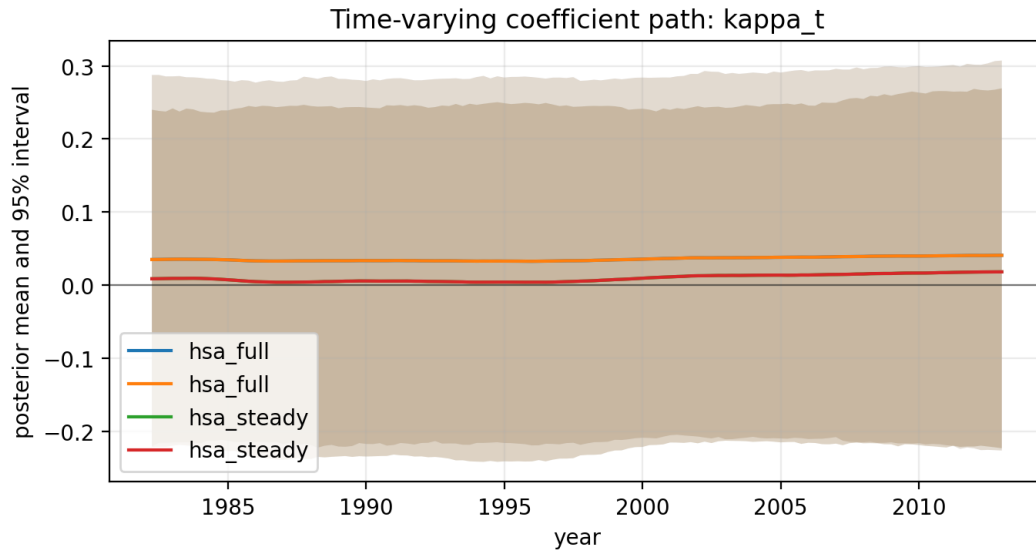
Prior vs posterior — hsa_full



6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.8721
ces	kappa=0	0.1000	0.2000	1.8721
hsa_dynamic	kappa=0	0.1000	0.2000	1.8766
hsa_dynamic	theta=0	0.1000	0.2000	0.3460
hsa_dynamic	kappa=0	0.1000	0.2000	1.8766
hsa_dynamic	theta=0	0.1000	0.2000	0.3460
hsa_full	kappa_0=0	0.1000	0.2000	1.7587
hsa_full	delta=0	0.0000	0.0200	0.9923
hsa_full	theta_0=0	0.1000	0.2000	0.4549
hsa_full	gamma=0	0.0000	0.0200	0.9571
hsa_full	kappa_0=0	0.1000	0.2000	1.7587
hsa_full	delta=0	0.0000	0.0200	0.9923
hsa_full	theta_0=0	0.1000	0.2000	0.4549
hsa_full	gamma=0	0.0000	0.0200	0.9571
hsa_steady	kappa_0=0	0.1000	0.2000	1.9878
hsa_steady	delta=0	0.0000	0.0200	1.0041
hsa_steady	kappa_0=0	0.1000	0.2000	1.9878
hsa_steady	delta=0	0.0000	0.0200	1.0041

6.4 Time-Varying Kappa Path



6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9350	0.6457	-225.9344	1.0000	NaN	NaN	NaN
ces	NaN	NaN	-225.9344	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.5602	0.6440	NaN	NaN	NaN	0.3460	NaN
hsa_dynamic	NaN	NaN	NaN	NaN	NaN	0.3460	NaN
hsa_full	-121.6984	0.6456	NaN	NaN	0.9923	NaN	0.9571
hsa_full	-121.6984	0.6456	NaN	NaN	0.9923	NaN	0.9571
hsa_steady	-121.9509	0.6457	-288.6606	0.0000	1.0041	NaN	NaN
hsa_steady	NaN	NaN	-288.6606	0.0000	1.0041	NaN	NaN

7 inv_markup: prior=baseline, period=full, constraint=kappa $\iota=0$

Condition. data=inv_markup, prior=baseline, period=full, constraint=restricted_kappa (kappa $\iota=0$).

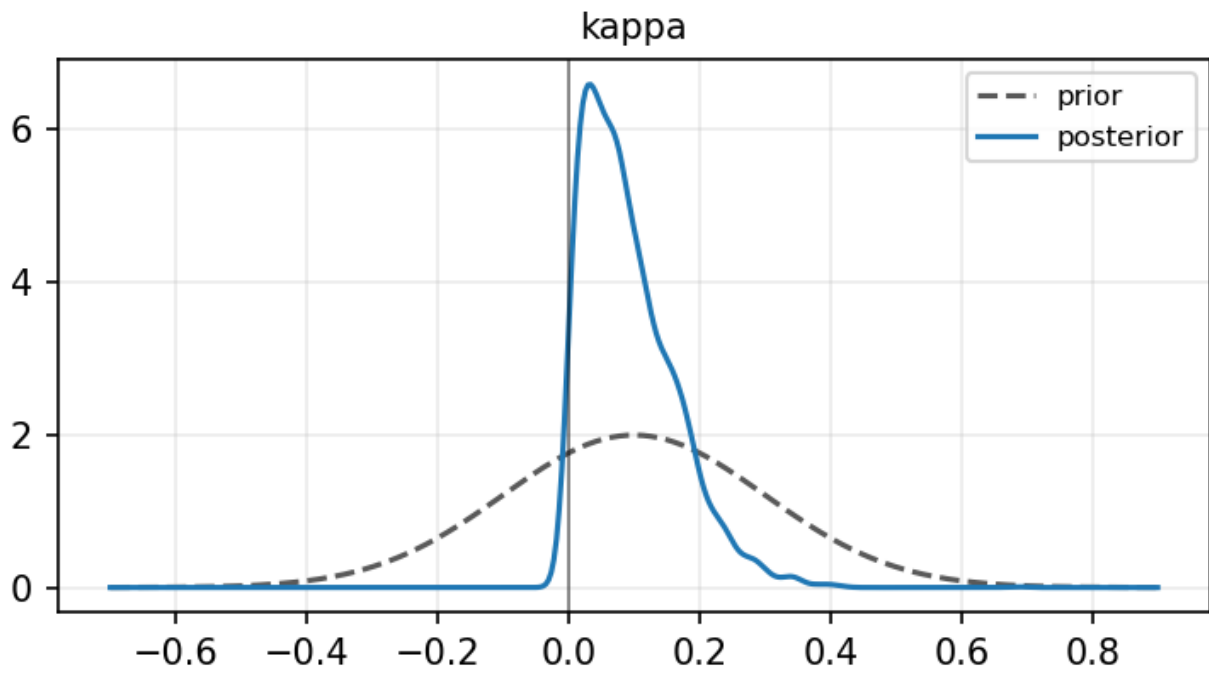
7.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.635 [0.533, 0.737]	0.639 [0.536, 0.742]	0.612 [0.498, 0.722]	0.602 [0.485, 0.718]
kappa	0.092 [0.004, 0.263]	NaN	0.093 [0.004, 0.263]	NaN
kappa_0	NaN	0.010 [-0.217, 0.241]	NaN	0.036 [-0.210, 0.283]
delta	NaN	-0.003 [-0.040, 0.034]	NaN	-0.002 [-0.040, 0.036]
theta	NaN	NaN	-0.001 [-0.363, 0.606]	NaN
theta_0	NaN	NaN	NaN	0.056 [-0.380, 0.472]
gamma	NaN	NaN	NaN	-0.006 [-0.046, 0.032]
rho_1	NaN	0.756 [0.389, 1.067]	0.899 [0.609, 1.112]	0.962 [0.725, 1.195]
rho_2	NaN	-0.022 [-0.313, 0.235]	0.014 [-0.191, 0.229]	-0.037 [-0.261, 0.184]
phi_1	0.662 [0.528, 0.794]	0.664 [0.536, 0.796]	0.664 [0.537, 0.789]	0.664 [0.529, 0.797]
lambda_ez	-0.096 [-0.377, 0.168]	-0.026 [-0.344, 0.288]	NaN	-0.036 [-0.344, 0.273]
n	NaN	-0.022 [-0.083, 0.040]	-0.033 [-0.081, 0.022]	-0.021 [-0.080, 0.041]

7.2 Prior vs Posterior by Model

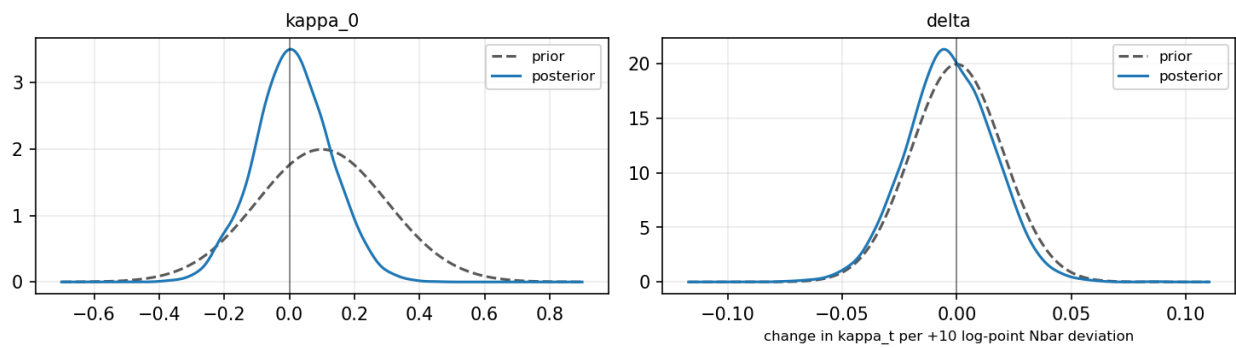
ces

Prior vs posterior — ces



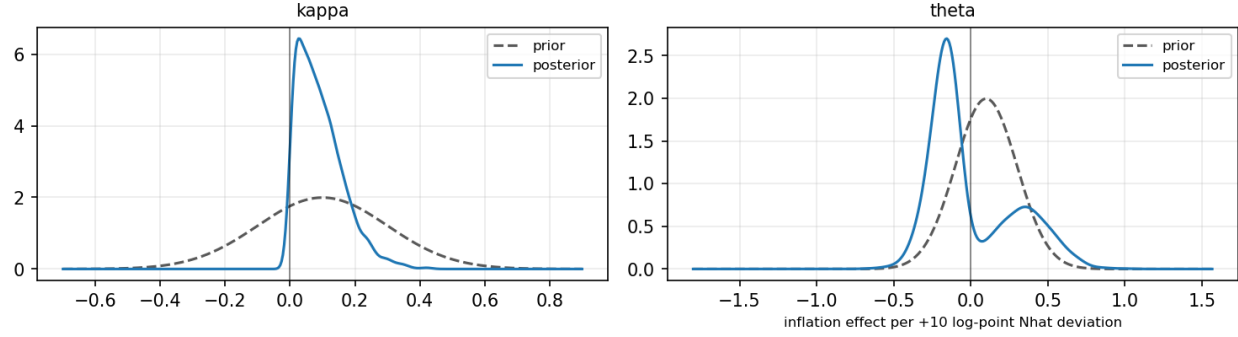
hsa_steady

Prior vs posterior — hsa_steady



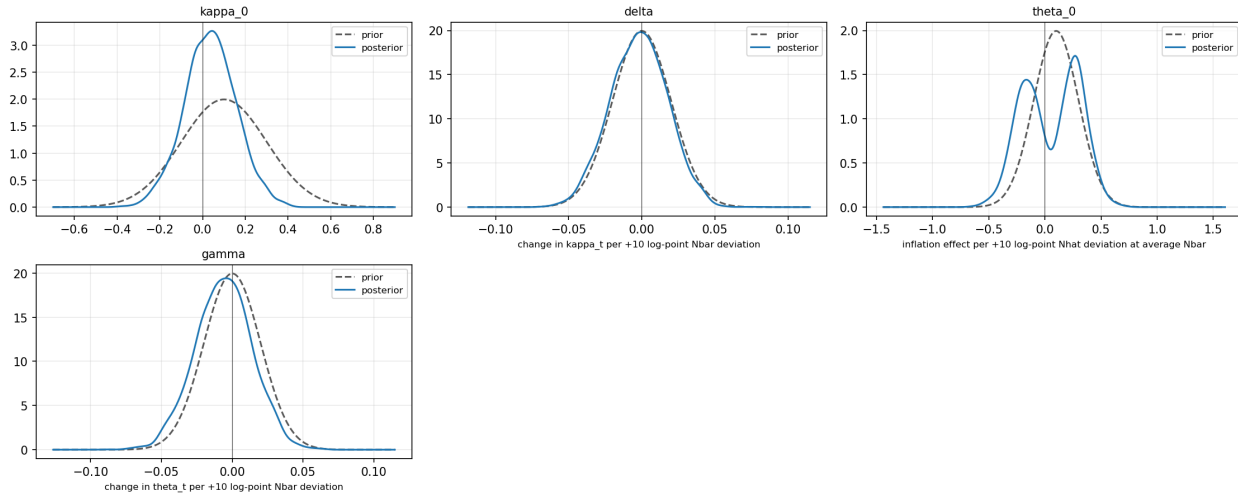
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

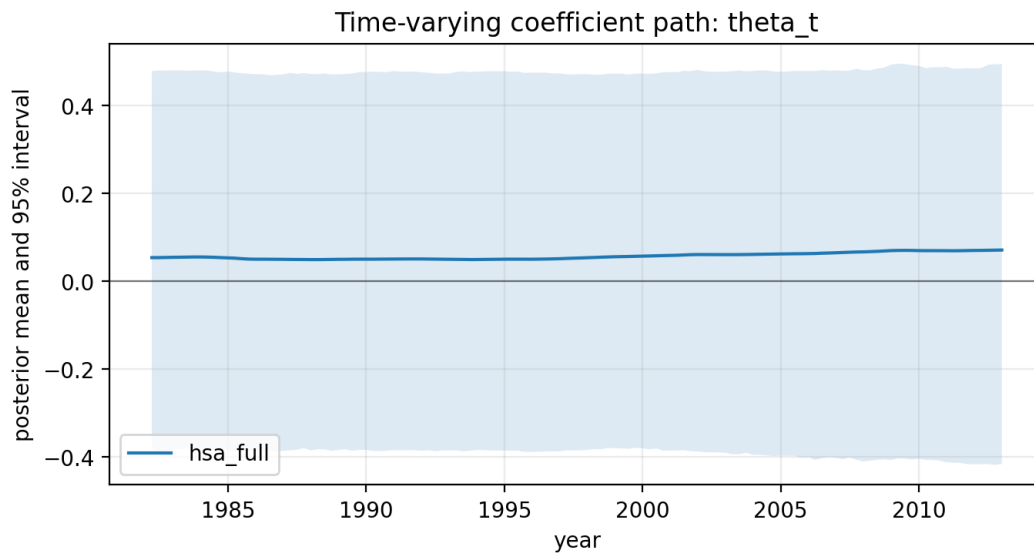
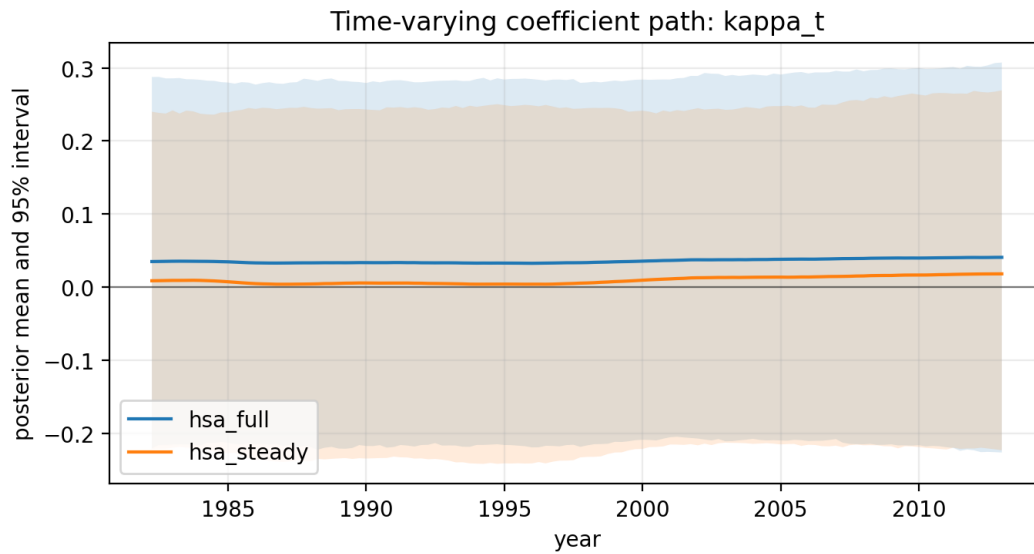
Prior vs posterior — hsa_full



7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.8901
hsa_dynamic	kappa=0	0.1000	0.2000	1.9811
hsa_dynamic	theta=0	0.1000	0.2000	0.3531
hsa_full	kappa_0=0	0.1000	0.2000	1.7587
hsa_full	delta=0	0.0000	0.0200	0.9923
hsa_full	theta_0=0	0.1000	0.2000	0.4549
hsa_full	gamma=0	0.0000	0.0200	0.9571
hsa_steady	kappa_0=0	0.1000	0.2000	1.9878
hsa_steady	delta=0	0.0000	0.0200	1.0041

7.4 Time-Varying Kappa Path



7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-122.6716	0.6498	-226.1735	1.0000	NaN	NaN	NaN
hsa_dynamic	-122.9043	0.6505	NaN	NaN	NaN	0.3531	NaN
hsa_full	-121.6984	0.6456	NaN	NaN	0.9923	NaN	0.9571
hsa_steady	-121.9509	0.6457	-288.6606	0.0000	1.0041	NaN	NaN

8 inv_markup: prior=baseline, period=pre_2008

Condition. data=inv_markup, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

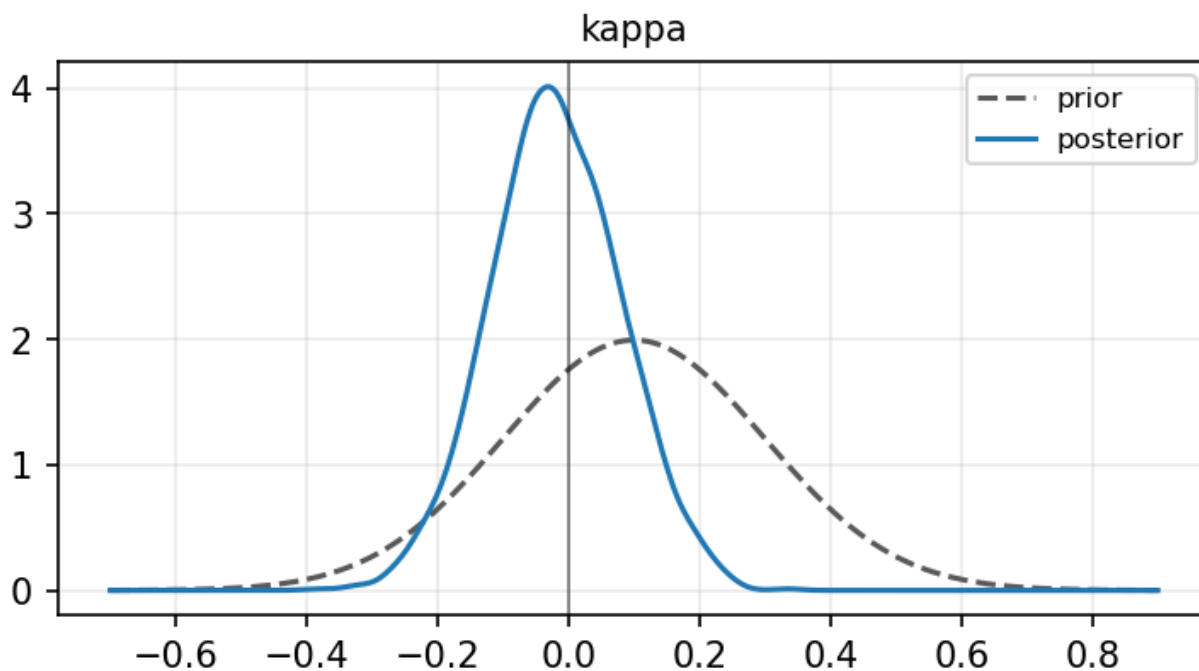
8.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.692 [0.581, 0.798]	0.680 [0.570, 0.744]	0.613 [0.484, 0.746]	0.590 [0.459, 0.720]
kappa	-0.020 [-0.216, 0.178]	NaN	0.036 [-0.177, 0.250]	NaN
kappa_0	NaN	0.127 [-0.264, 0.579]	NaN	0.070 [-0.143, 0.281]
delta	NaN	-0.001 [-0.003, 0.001]	NaN	-0.004 [-0.042, 0.034]
theta	NaN	NaN	0.346 [-0.340, 0.756]	NaN
theta_0	NaN	NaN	NaN	0.356 [0.148, 0.606]
gamma	NaN	NaN	NaN	-0.003 [-0.042, 0.033]
rho_1	NaN	1.811 [1.702, 1.891]	0.893 [0.656, 1.121]	0.959 [0.714, 1.206]
rho_2	NaN	-0.816 [-0.894, -0.707]	0.005 [-0.198, 0.214]	-0.050 [-0.287, 0.178]
phi_1	0.712 [0.584, 0.840]	0.706 [0.613, 0.792]	0.709 [0.579, 0.835]	0.712 [0.575, 0.840]
lambda_ez	0.014 [-0.285, 0.307]	0.073 [-0.219, 0.349]	NaN	-0.009 [-0.280, 0.273]
n	NaN	0.011 [-0.149, 0.137]	-0.009 [-0.072, 0.044]	0.002 [-0.060, 0.065]

8.2 Prior vs Posterior by Model

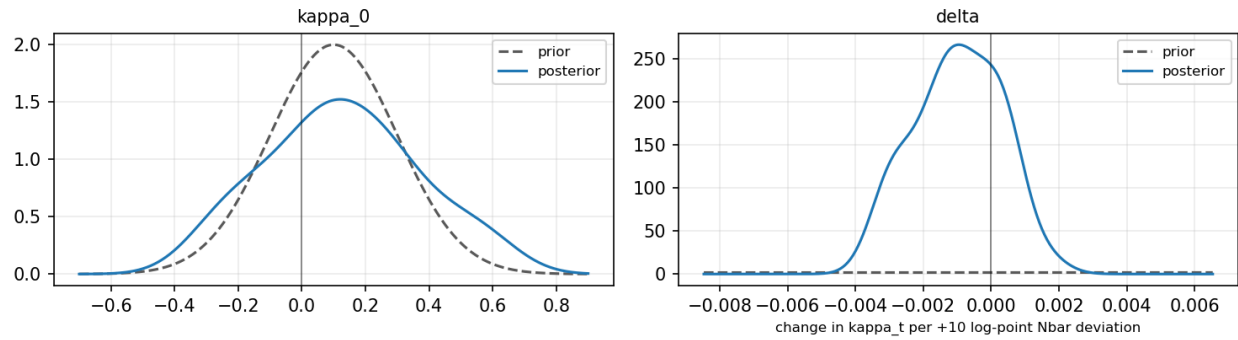
ces

Prior vs posterior — ces



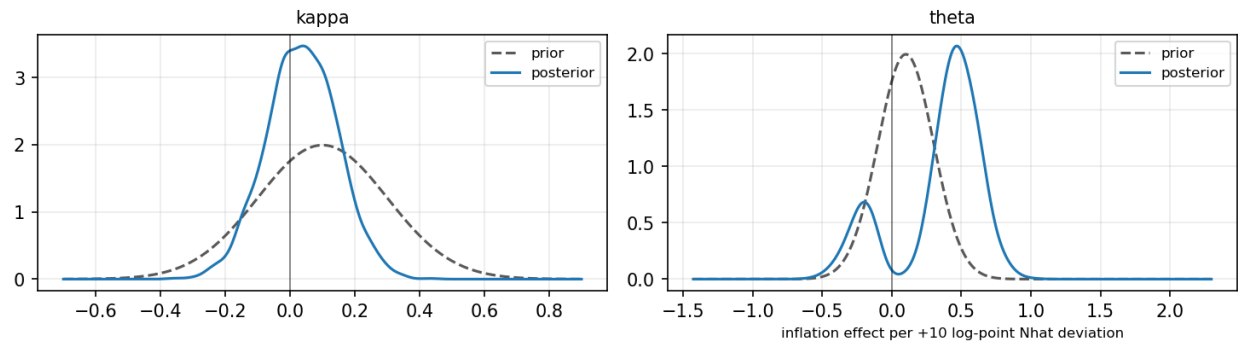
hsa_steady

Prior vs posterior — hsa_steady



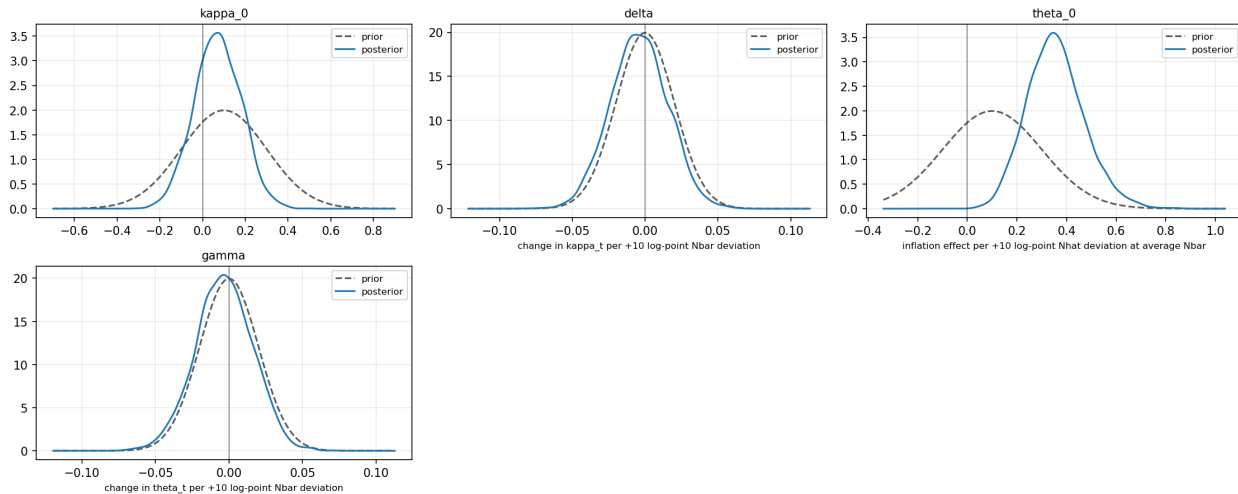
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

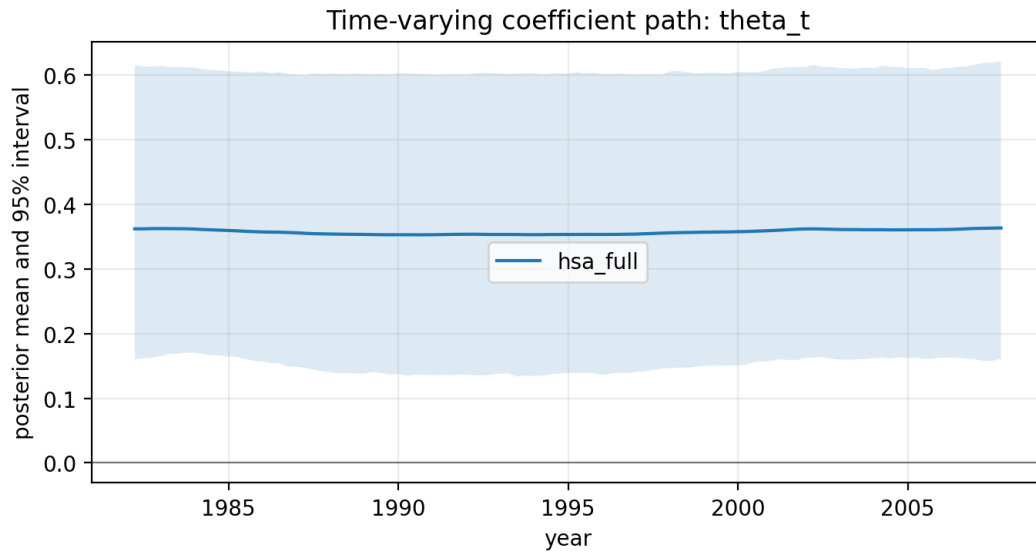
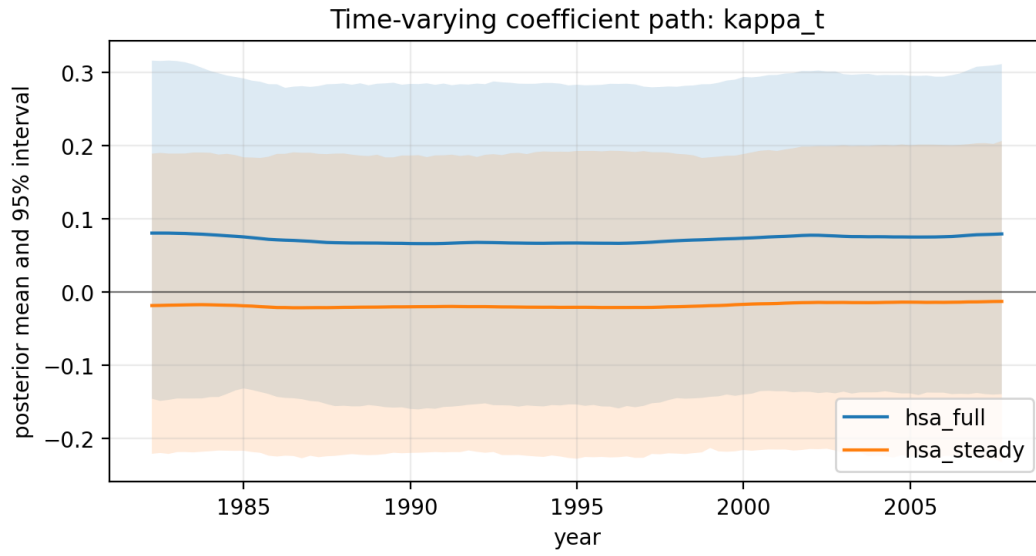
Prior vs posterior — hsa_full



8.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.1324
hsa_dynamic	kappa=0	0.1000	0.2000	1.9324
hsa_dynamic	theta=0	0.1000	0.2000	0.0449
hsa_full	kappa_0=0	0.1000	0.2000	1.7135
hsa_full	delta=0	0.0000	0.0200	0.9750
hsa_full	theta_0=0	0.1000	0.2000	0.0024
hsa_full	gamma=0	0.0000	0.0200	1.0022
hsa_steady	kappa_0=0	0.1000	0.2000	2.1685
hsa_steady	delta=0	0.0000	0.0200	0.9608

8.4 Time-Varying Kappa Path



8.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-78.1989	0.5141	-155.3906	1.0000	NaN	NaN	NaN
hsa_dynamic	-67.3066	0.4643	NaN	NaN	NaN	0.0449	NaN
hsa_full	-57.2296	0.4125	NaN	NaN	0.9750	NaN	1.0022
hsa_steady	-78.1529	0.5139	-217.5896	0.0000	0.9608	NaN	NaN

9 inv_markup: prior=baseline, period=post_2008

Condition. data=inv_markup, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

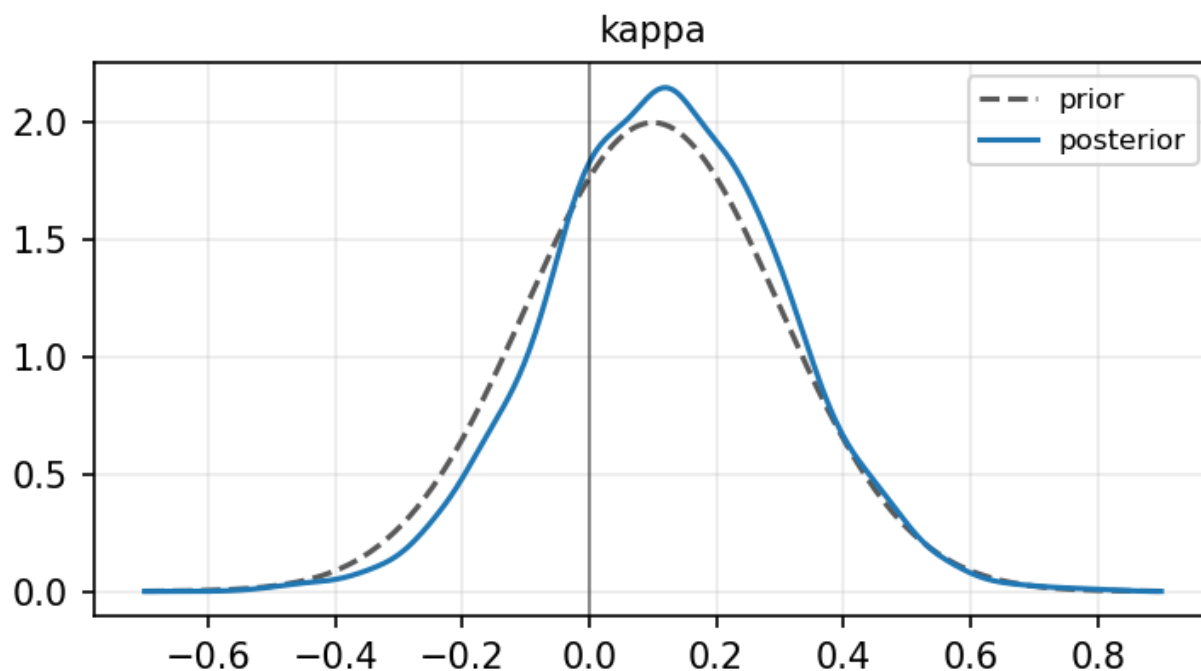
9.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.568 [0.322, 0.807]	0.568 [0.372, 0.797]	0.575 [0.347, 0.793]	0.571 [0.330, 0.803]
kappa	0.122 [-0.240, 0.481]	NaN	0.192 [-0.180, 0.562]	NaN
kappa_0	NaN	0.129 [-0.275, 0.614]	NaN	0.124 [-0.215, 0.481]
delta	NaN	0.001 [-0.004, 0.006]	NaN	0.000 [-0.038, 0.041]
theta	NaN	NaN	0.164 [-0.316, 0.637]	NaN
theta_0	NaN	NaN	NaN	0.100 [-0.316, 0.503]
gamma	NaN	NaN	NaN	-0.000 [-0.039, 0.039]
rho_1	NaN	0.514 [0.247, 0.812]	0.534 [0.137, 0.910]	0.575 [0.178, 0.961]
rho_2	NaN	-0.453 [-0.715, -0.197]	-0.346 [-0.717, 0.002]	-0.361 [-0.723, 0.009]
phi_1	0.627 [0.322, 0.944]	0.643 [0.421, 0.885]	0.596 [0.314, 0.892]	0.624 [0.308, 0.939]
lambda_ez	0.039 [-0.560, 0.621]	-0.146 [-0.825, 0.527]	NaN	0.019 [-0.568, 0.597]
n	NaN	-0.071 [-0.218, 0.178]	-0.007 [-0.164, 0.147]	-0.002 [-0.174, 0.174]

9.2 Prior vs Posterior by Model

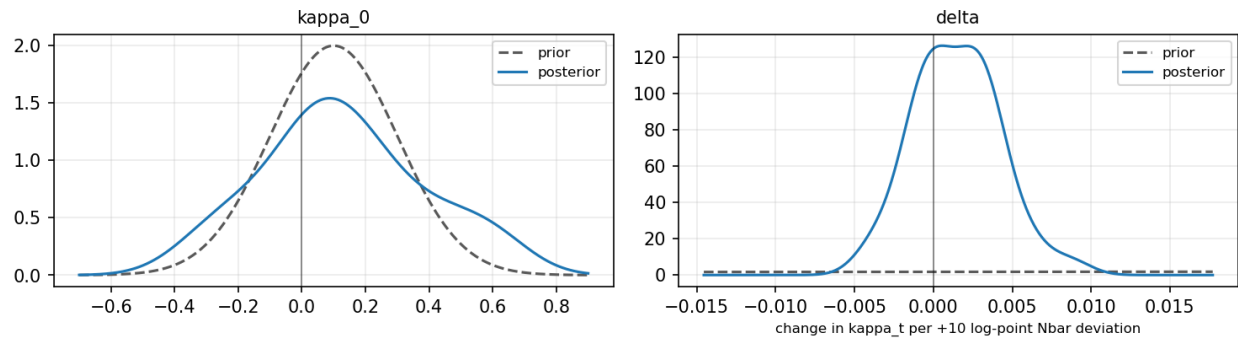
ces

Prior vs posterior — ces



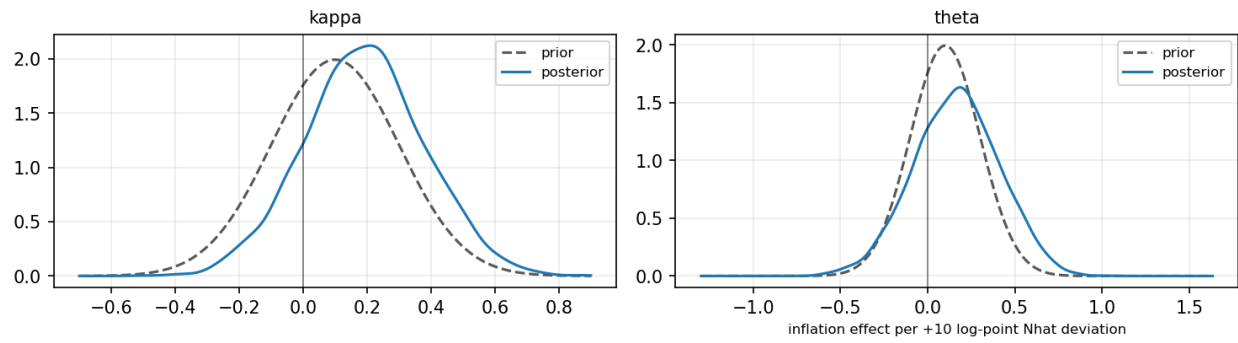
hsa_steady

Prior vs posterior — hsa_steady



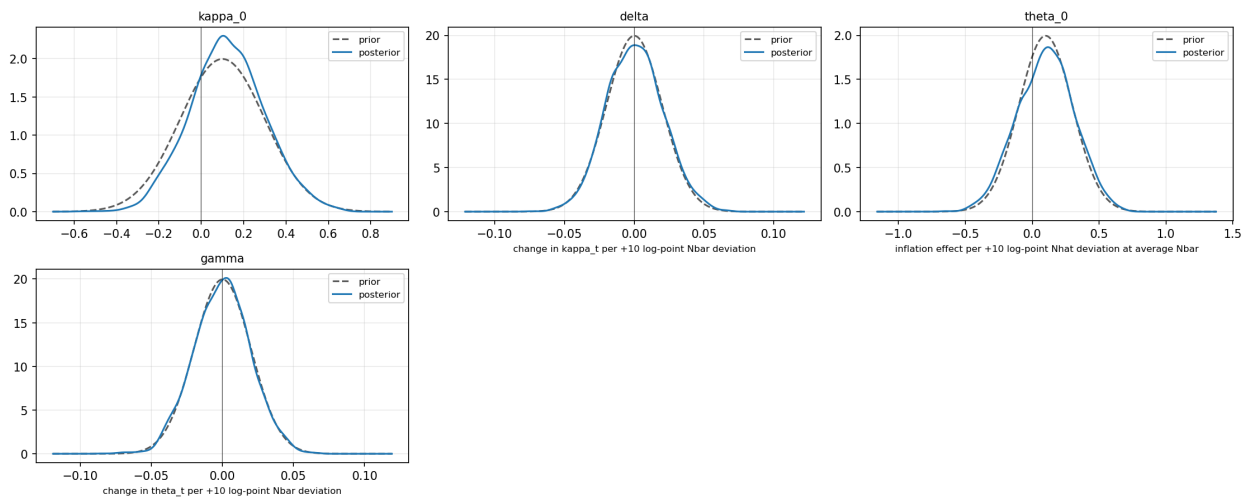
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

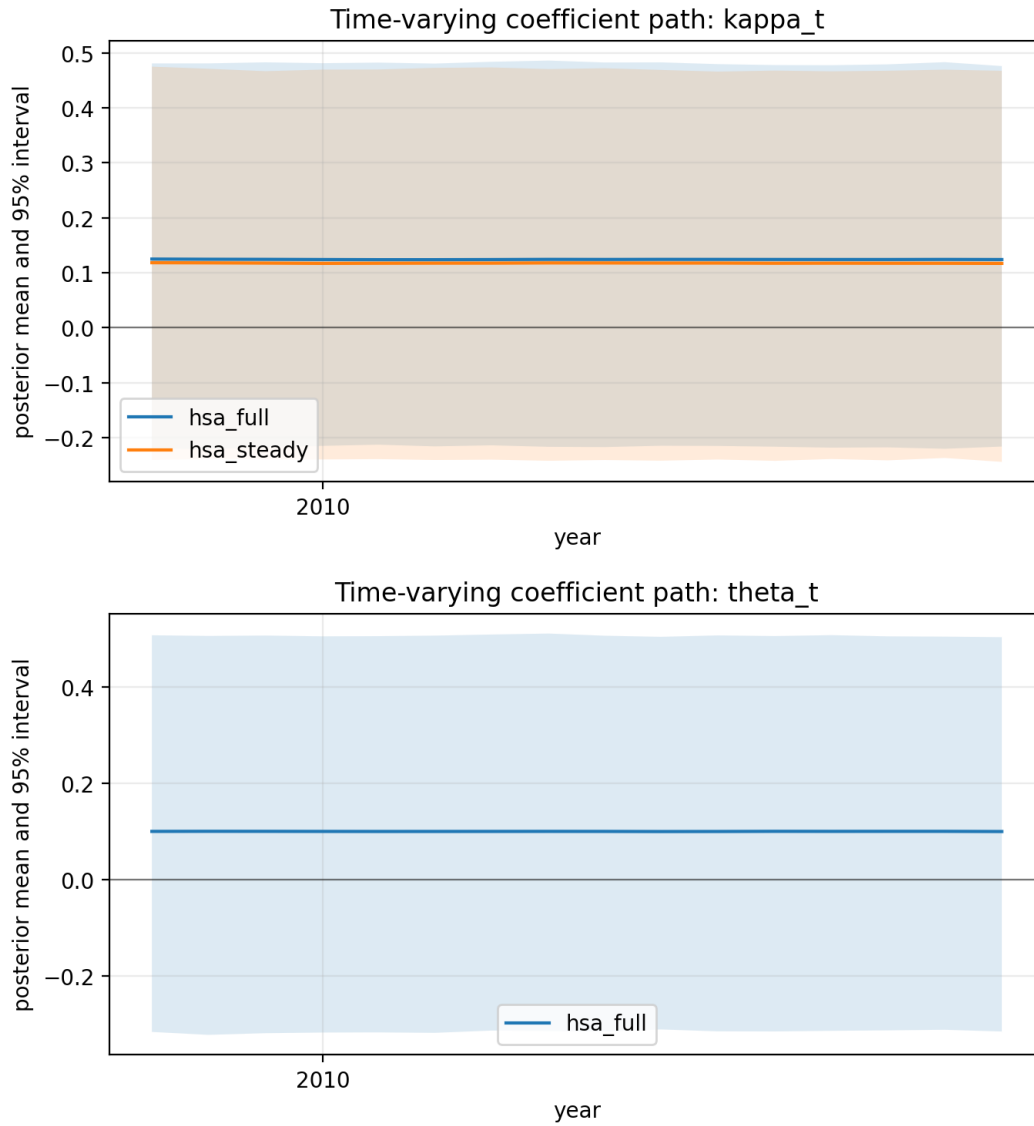
Prior vs posterior — hsa_full



9.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.0368
hsa_dynamic	kappa=0	0.1000	0.2000	0.6939
hsa_dynamic	theta=0	0.1000	0.2000	0.7287
hsa_full	kappa_0=0	0.1000	0.2000	1.0178
hsa_full	delta=0	0.0000	0.0200	0.9467
hsa_full	theta_0=0	0.1000	0.2000	0.8543
hsa_full	gamma=0	0.0000	0.0200	0.9970
hsa_steady	kappa_0=0	0.1000	0.2000	1.0301
hsa_steady	delta=0	0.0000	0.0200	0.9934

9.4 Time-Varying Kappa Path



9.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-21.5563	0.9177	-48.8693	1.0000	NaN	NaN	NaN
hsa_dynamic	-21.0155	0.8974	NaN	NaN	NaN	0.7287	NaN
hsa_full	-21.4022	0.9116	NaN	NaN	0.9467	NaN	0.9970
hsa_steady	-21.5369	0.9168	-72.4978	0.0000	0.9934	NaN	NaN

10 inv_markup: prior=baseline, period=exclude_covid

Condition. data=inv_markup, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

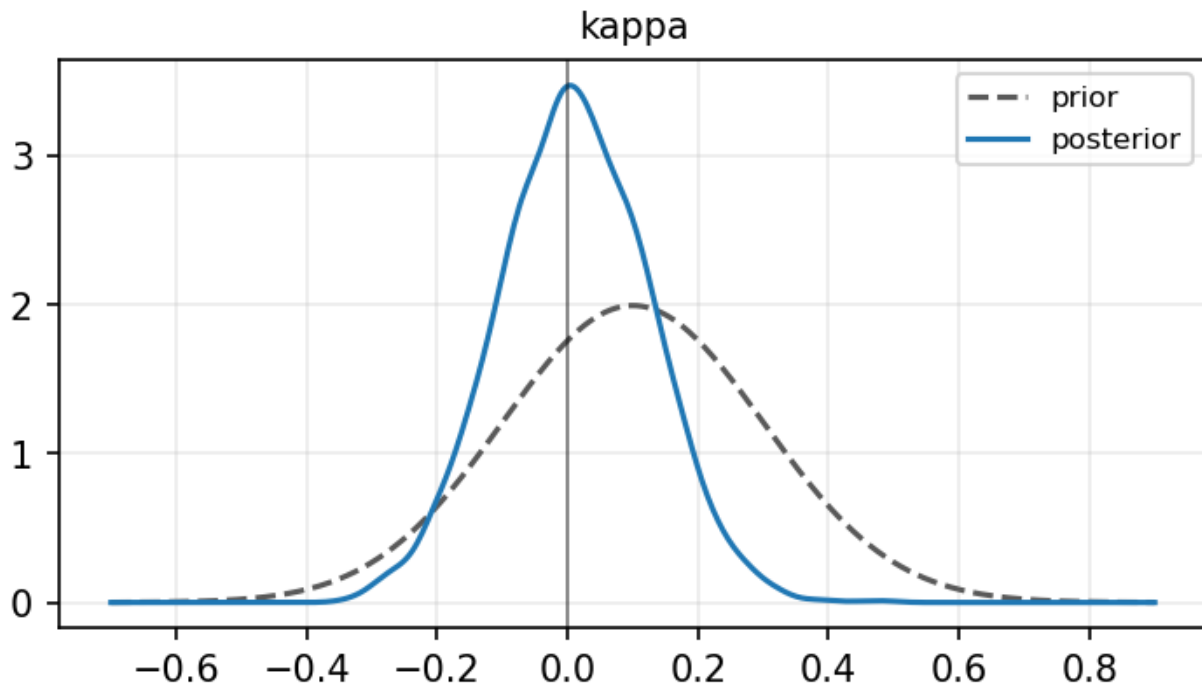
10.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.638 [0.531, 0.741]	0.635 [0.550, 0.733]	0.617 [0.506, 0.721]	0.576 [0.464, 0.689]
kappa	0.011 [-0.212, 0.234]	NaN	0.009 [-0.220, 0.247]	NaN
kappa_0	NaN	0.087 [-0.223, 0.558]	NaN	0.075 [-0.158, 0.319]
delta	NaN	-0.000 [-0.003, 0.002]	NaN	-0.004 [-0.042, 0.034]
theta	NaN	NaN	-0.022 [-0.368, 0.573]	NaN
theta_0	NaN	NaN	NaN	0.316 [0.103, 0.588]
gamma	NaN	NaN	NaN	-0.004 [-0.040, 0.034]
rho_1	NaN	1.831 [1.731, 1.922]	0.902 [0.628, 1.116]	0.960 [0.726, 1.183]
rho_2	NaN	-0.833 [-0.928, -0.736]	0.010 [-0.195, 0.225]	-0.041 [-0.262, 0.170]
phi_1	0.665 [0.535, 0.796]	0.663 [0.545, 0.783]	0.667 [0.541, 0.793]	0.665 [0.534, 0.794]
lambda_ez	-0.021 [-0.334, 0.287]	-0.063 [-0.360, 0.384]	NaN	-0.047 [-0.354, 0.259]
n	NaN	0.013 [-0.147, 0.201]	-0.033 [-0.081, 0.020]	-0.006 [-0.059, 0.045]

10.2 Prior vs Posterior by Model

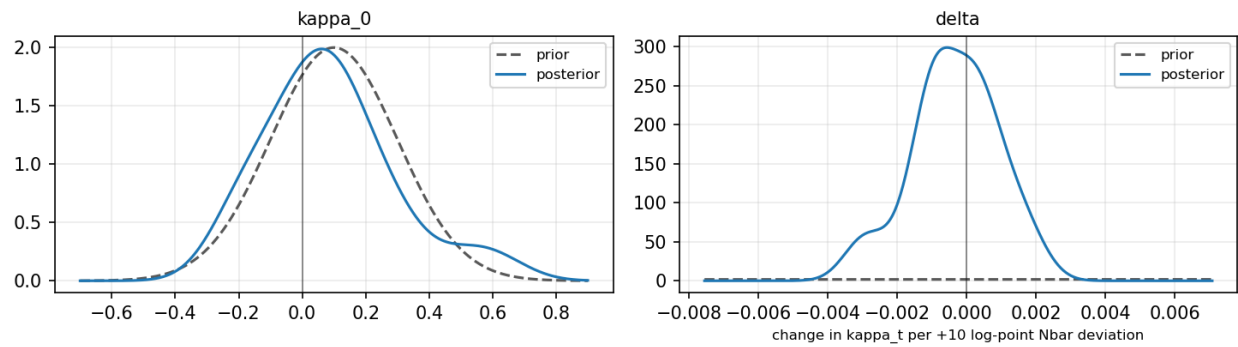
ces

Prior vs posterior — ces



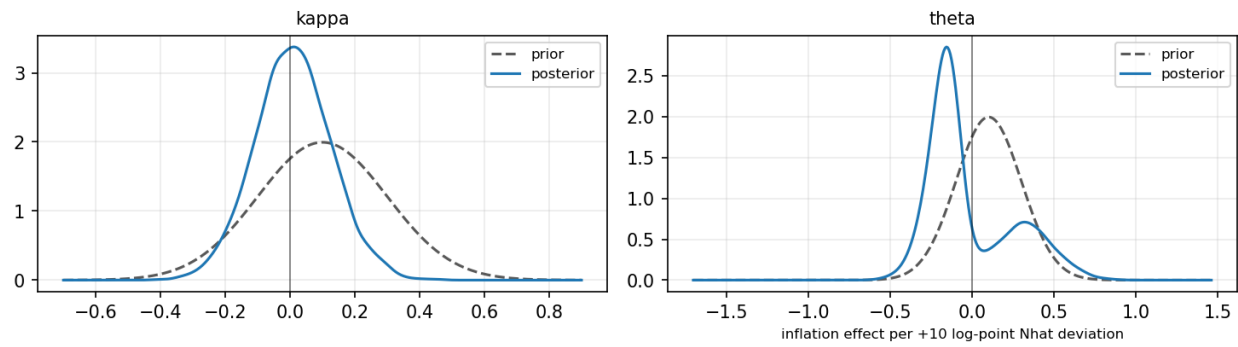
hsa_steady

Prior vs posterior — hsa_steady



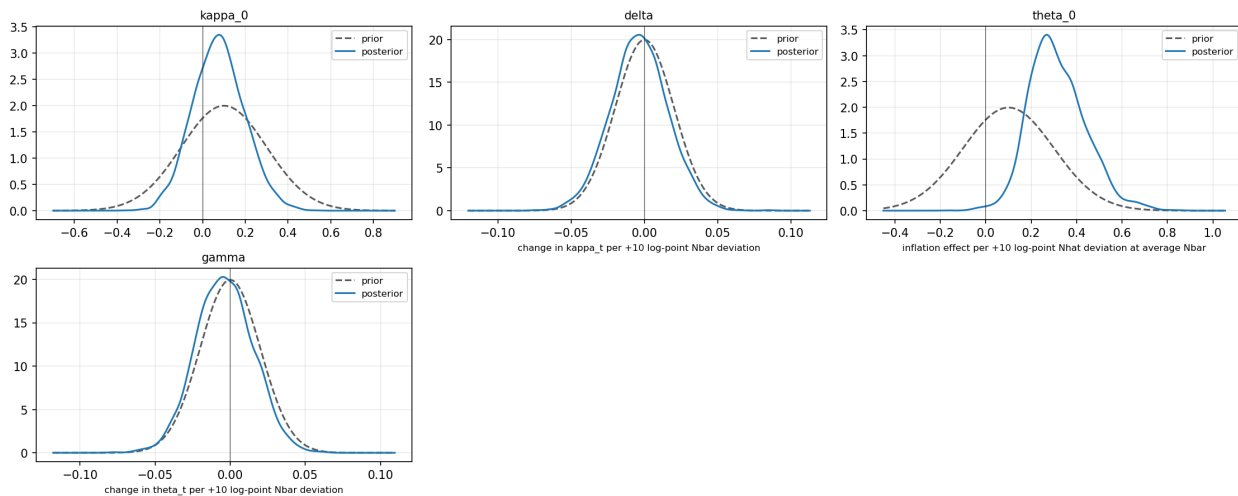
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

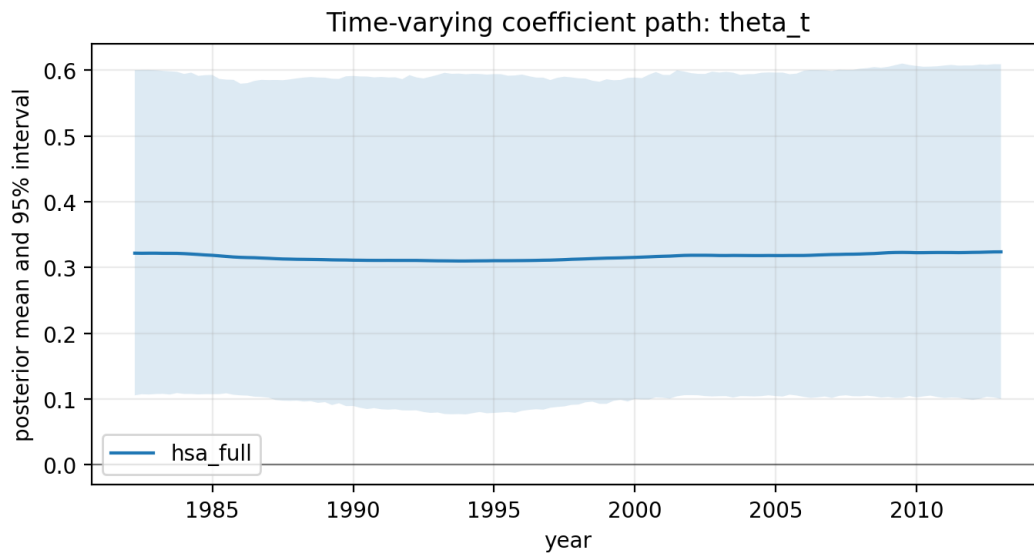
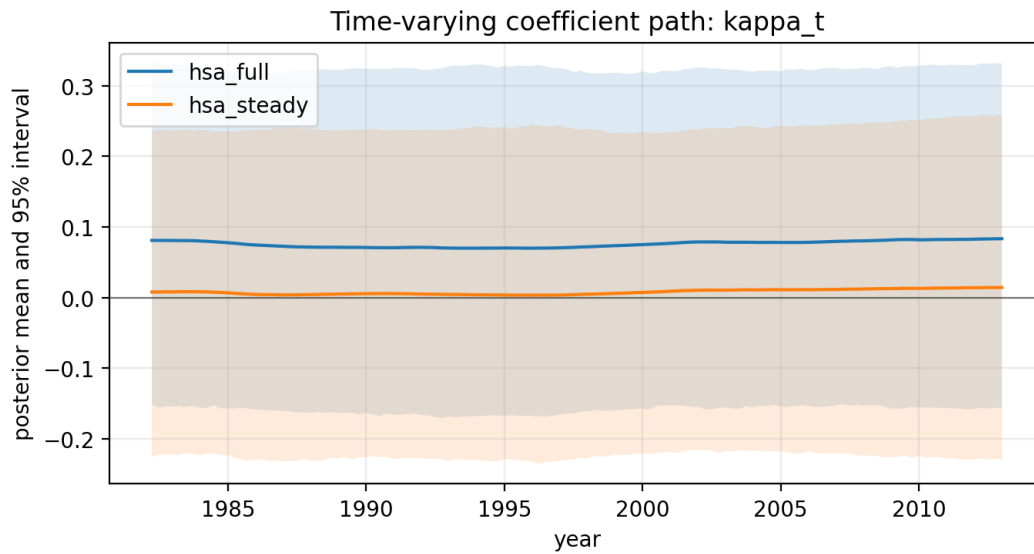
Prior vs posterior — hsa_full



10.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.9711
hsa_dynamic	kappa=0	0.1000	0.2000	1.9085
hsa_dynamic	theta=0	0.1000	0.2000	0.3537
hsa_full	kappa_0=0	0.1000	0.2000	1.5391
hsa_full	delta=0	0.0000	0.0200	1.0075
hsa_full	theta_0=0	0.1000	0.2000	0.0477
hsa_full	gamma=0	0.0000	0.0200	0.9916
hsa_steady	kappa_0=0	0.1000	0.2000	1.9859
hsa_steady	delta=0	0.0000	0.0200	1.0763

10.4 Time-Varying Kappa Path



10.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9854	0.6458	-225.9372	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.4893	0.6438	NaN	NaN	NaN	0.3537	NaN
hsa_full	-110.0366	0.5844	NaN	NaN	1.0075	NaN	0.9916
hsa_steady	-121.9323	0.6457	-288.8480	0.0000	1.0763	NaN	NaN

11 inv_markup: prior=baseline, period=start_1988

Condition. data=inv_markup, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

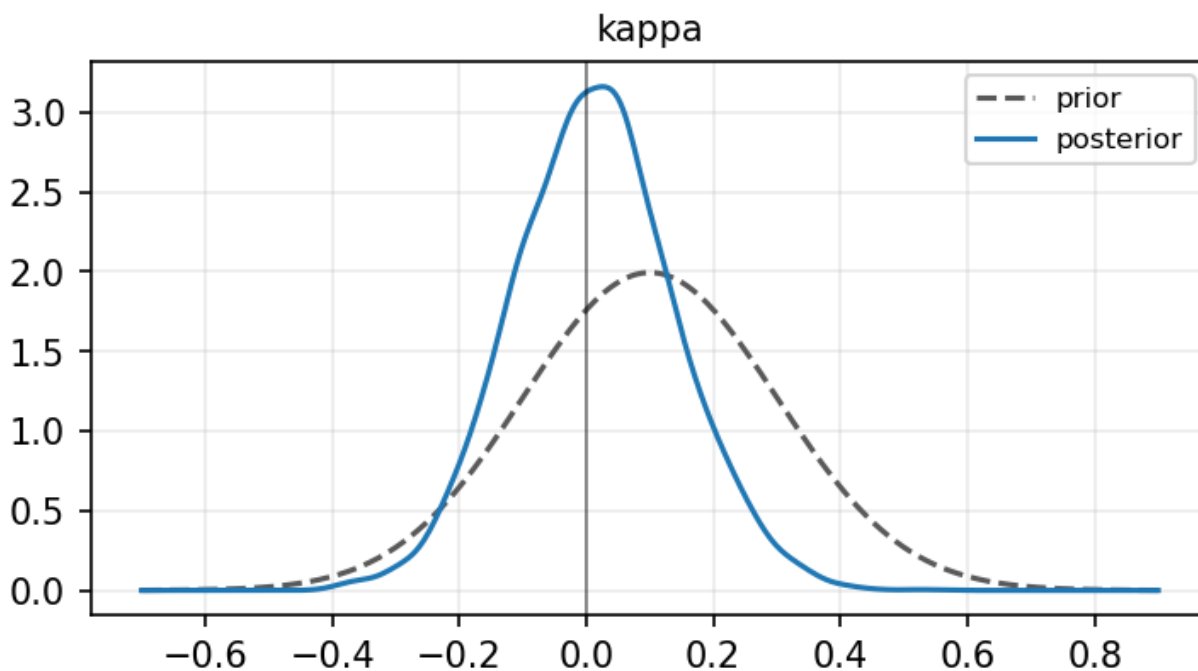
11.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.622 [0.497, 0.742]	0.619 [0.522, 0.715]	0.612 [0.494, 0.724]	0.580 [0.447, 0.715]
kappa	0.012 [-0.232, 0.268]	NaN	0.020 [-0.225, 0.273]	NaN
kappa_0	NaN	0.069 [-0.271, 0.412]	NaN	0.059 [-0.199, 0.319]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	-0.001 [-0.038, 0.037]
theta	NaN	NaN	0.145 [-0.247, 0.525]	NaN
theta_0	NaN	NaN	NaN	0.157 [-0.115, 0.423]
gamma	NaN	NaN	NaN	-0.001 [-0.040, 0.037]
rho_1	NaN	1.739 [1.593, 1.860]	0.669 [0.266, 1.038]	0.959 [0.728, 1.202]
rho_2	NaN	-0.743 [-0.863, -0.594]	-0.074 [-0.378, 0.197]	-0.038 [-0.261, 0.188]
phi_1	0.619 [0.468, 0.772]	0.615 [0.497, 0.728]	0.622 [0.466, 0.774]	0.616 [0.461, 0.771]
lambda_ez	-0.006 [-0.340, 0.325]	0.022 [-0.371, 0.432]	NaN	-0.030 [-0.371, 0.299]
n	NaN	-0.194 [-0.364, -0.075]	-0.042 [-0.087, 0.003]	-0.050 [-0.113, 0.013]

11.2 Prior vs Posterior by Model

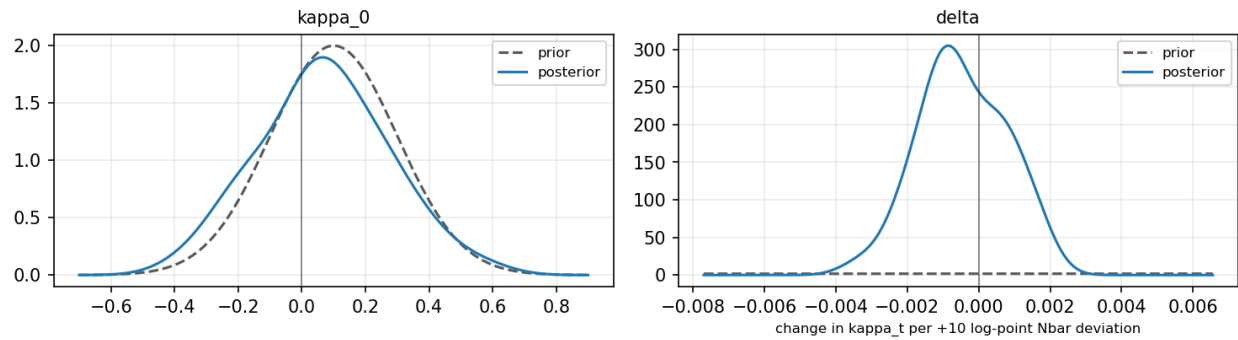
ces

Prior vs posterior — ces



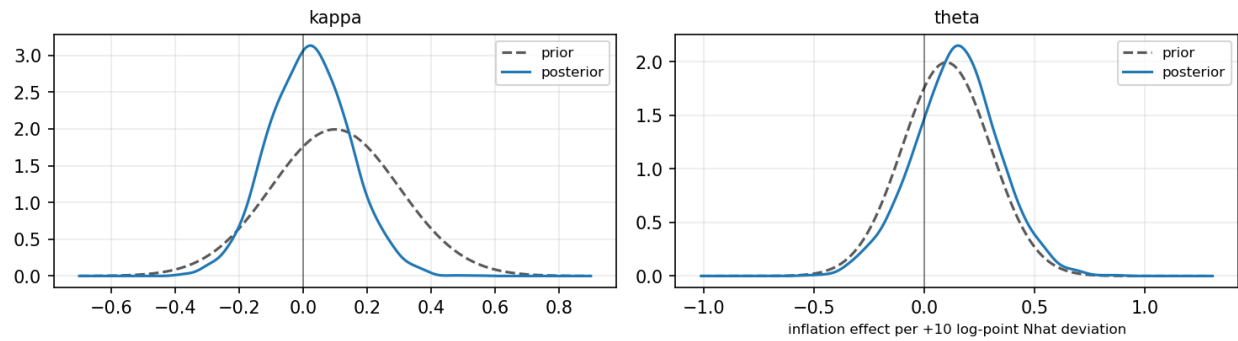
hsa_steady

Prior vs posterior — hsa_steady



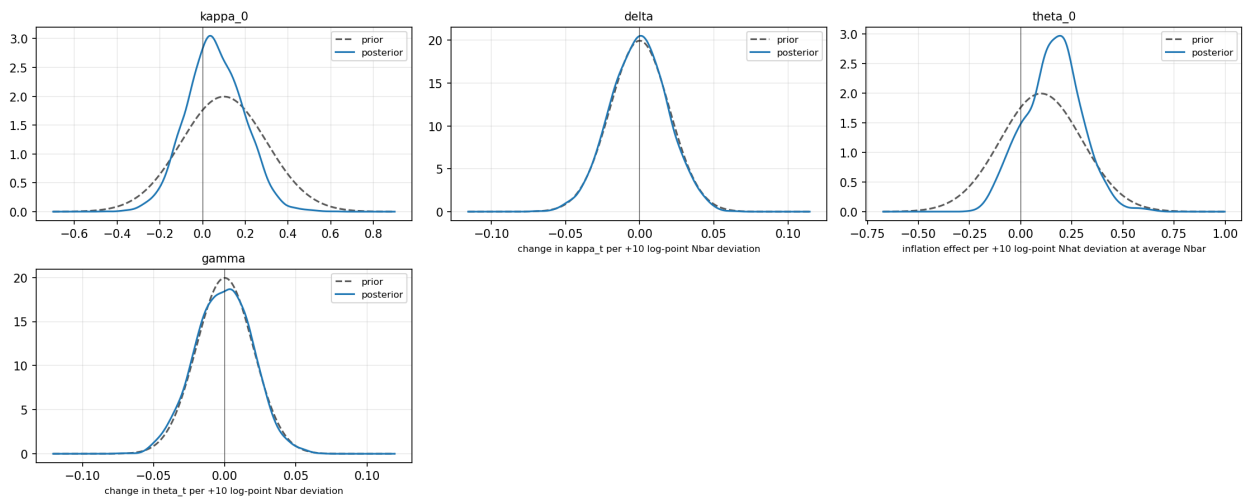
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

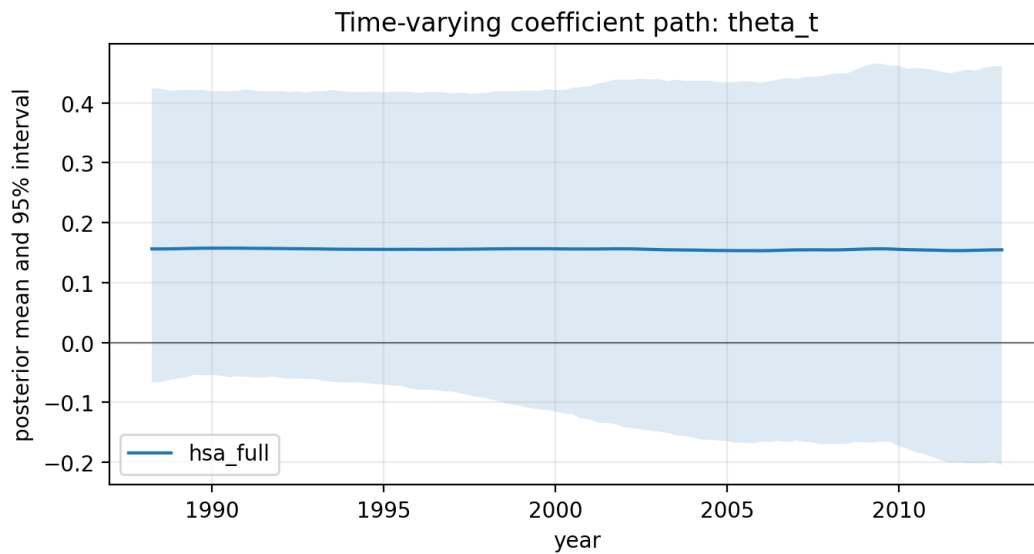
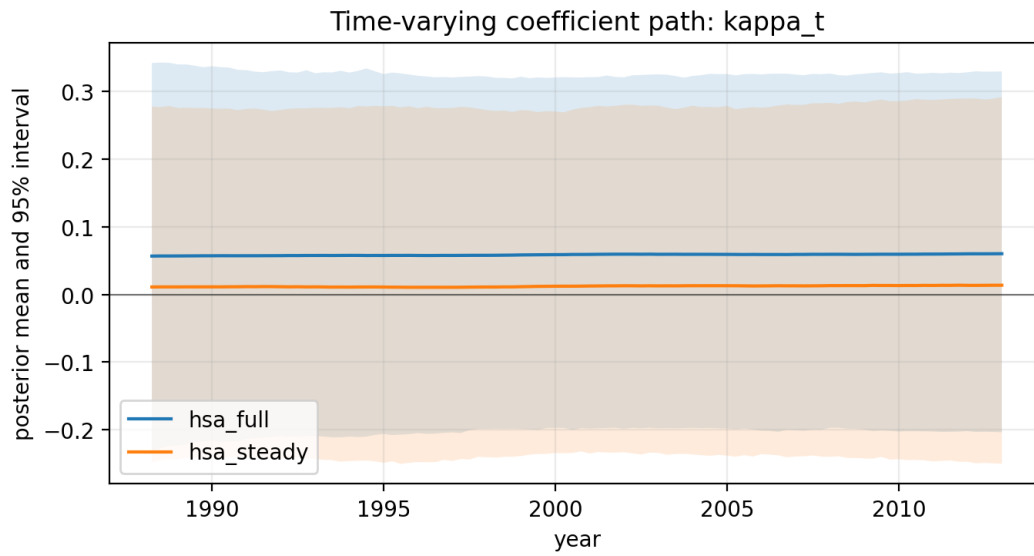
Prior vs posterior — hsa_full



11.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.7767
hsa_dynamic	kappa=0	0.1000	0.2000	1.7400
hsa_dynamic	theta=0	0.1000	0.2000	0.8377
hsa_full	kappa_0=0	0.1000	0.2000	1.6045
hsa_full	delta=0	0.0000	0.0200	1.0279
hsa_full	theta_0=0	0.1000	0.2000	0.8423
hsa_full	gamma=0	0.0000	0.0200	0.9246
hsa_steady	kappa_0=0	0.1000	0.2000	1.7121
hsa_steady	delta=0	0.0000	0.0200	0.9776

11.4 Time-Varying Kappa Path



11.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-94.9552	0.6233	-184.8106	1.0000	NaN	NaN	NaN
hsa_dynamic	-93.7620	0.6179	NaN	NaN	NaN	0.8377	NaN
hsa_full	-92.3101	0.6067	NaN	NaN	1.0279	NaN	0.9246
hsa_steady	-94.9375	0.6233	-244.5728	0.0000	0.9776	NaN	NaN

12 inv_markup: prior=baseline, period=end_2019

Condition. data=inv_markup, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

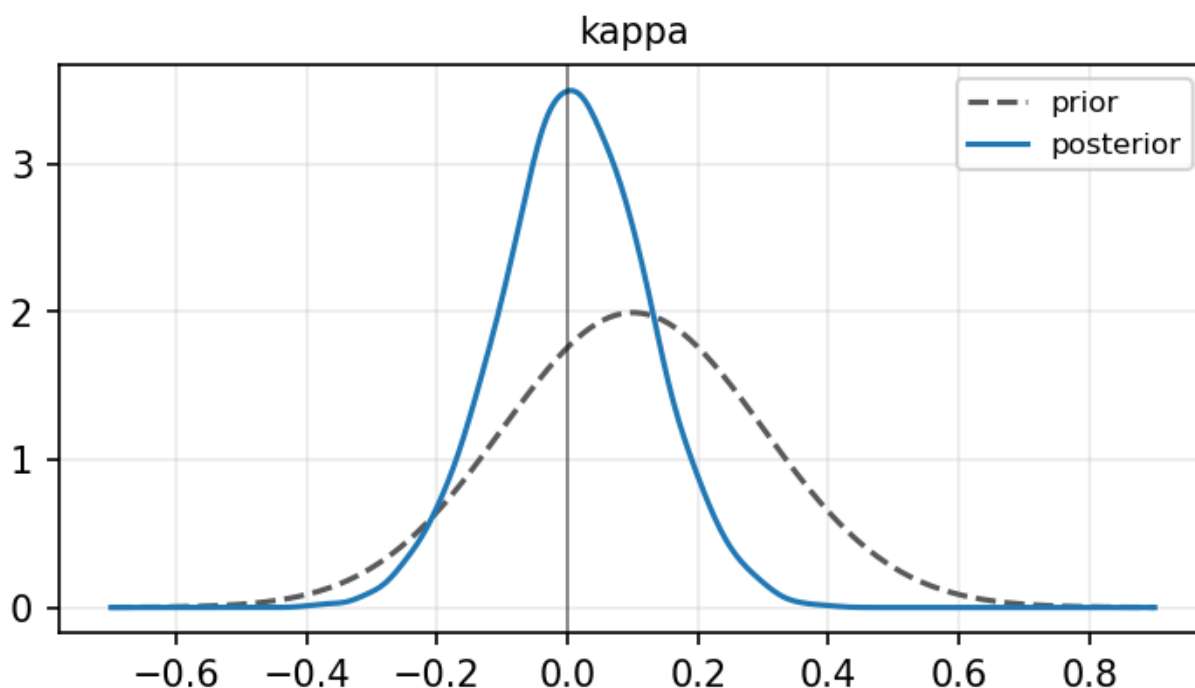
12.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.639 [0.537, 0.740]	0.646 [0.551, 0.726]	0.617 [0.505, 0.719]	0.628 [0.524, 0.730]
kappa	0.009 [-0.217, 0.231]	NaN	0.008 [-0.224, 0.237]	NaN
kappa_0	NaN	0.128 [-0.243, 0.541]	NaN	0.007 [-0.219, 0.226]
delta	NaN	-0.001 [-0.004, 0.002]	NaN	-0.003 [-0.039, 0.033]
theta	NaN	NaN	-0.039 [-0.354, 0.565]	NaN
theta_0	NaN	NaN	NaN	-0.165 [-0.391, 0.043]
gamma	NaN	NaN	NaN	-0.004 [-0.041, 0.035]
rho_1	NaN	1.807 [1.740, 1.871]	0.903 [0.630, 1.106]	0.929 [0.698, 1.161]
rho_2	NaN	-0.811 [-0.876, -0.747]	0.014 [-0.176, 0.236]	-0.017 [-0.232, 0.196]
phi_1	0.665 [0.534, 0.789]	0.664 [0.476, 0.814]	0.664 [0.541, 0.794]	0.665 [0.536, 0.794]
lambda_ez	-0.022 [-0.336, 0.295]	0.027 [-0.270, 0.304]	NaN	-0.031 [-0.339, 0.270]
n	NaN	-0.048 [-0.153, 0.048]	-0.034 [-0.083, 0.021]	-0.042 [-0.093, 0.010]

12.2 Prior vs Posterior by Model

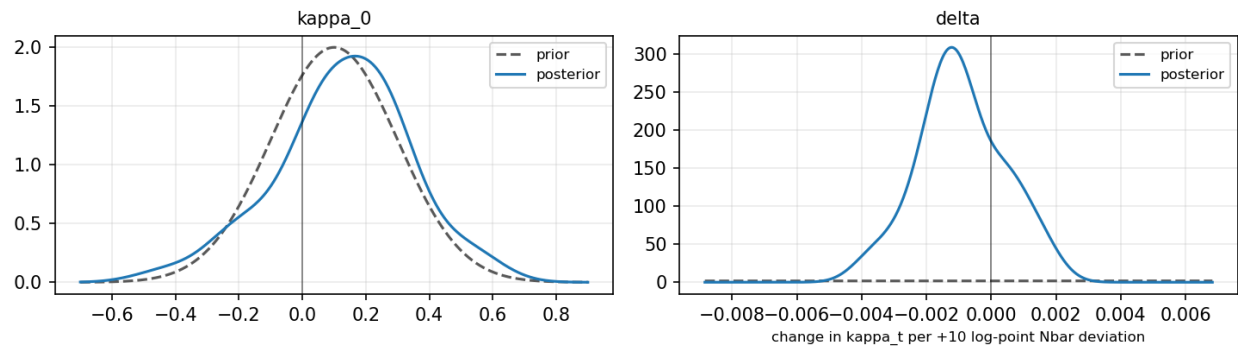
ces

Prior vs posterior — ces



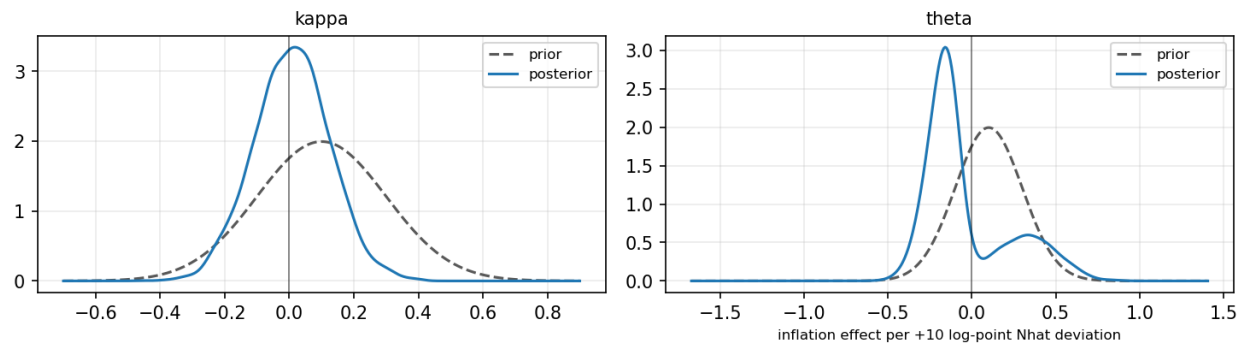
hsa_steady

Prior vs posterior — hsa_steady



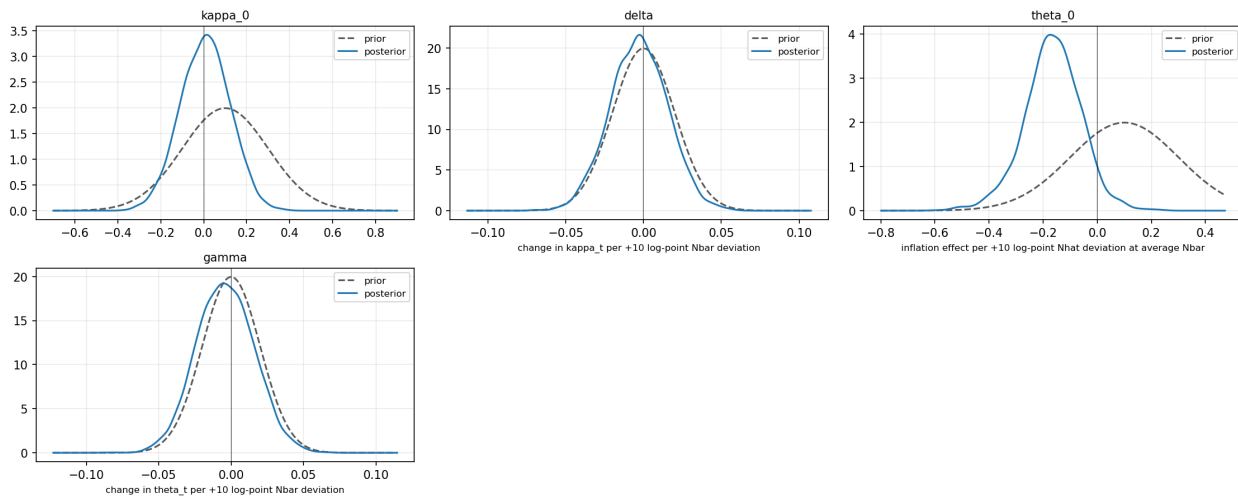
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

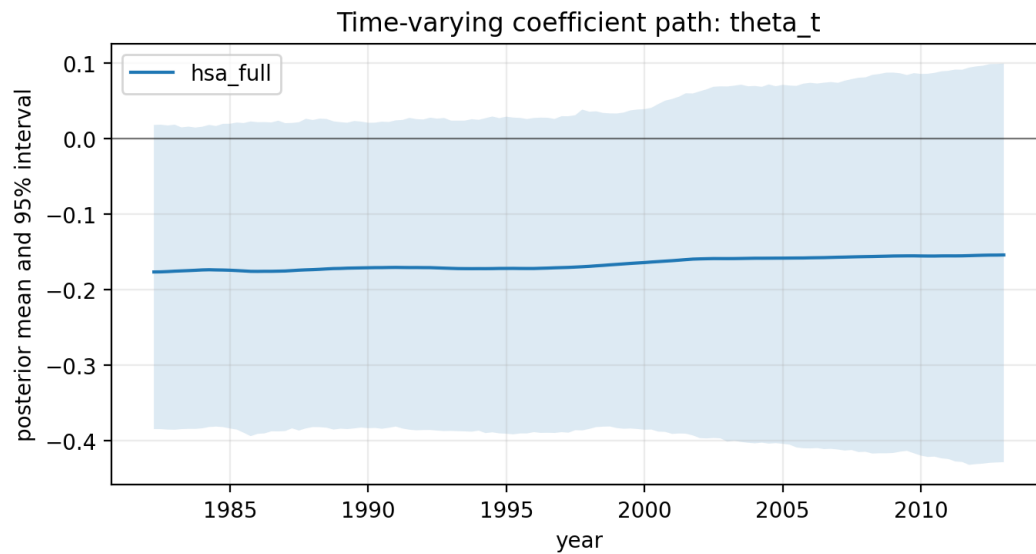
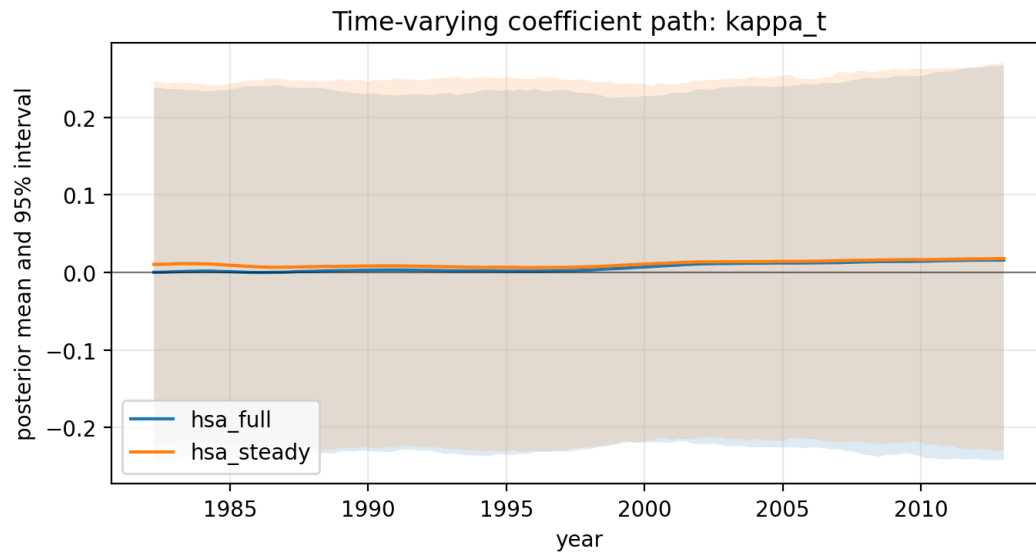
Prior vs posterior — hsa_full



12.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.9847
hsa_dynamic	kappa=0	0.1000	0.2000	1.8803
hsa_dynamic	theta=0	0.1000	0.2000	0.3289
hsa_full	kappa_0=0	0.1000	0.2000	1.9236
hsa_full	delta=0	0.0000	0.0200	1.0576
hsa_full	theta_0=0	0.1000	0.2000	0.5718
hsa_full	gamma=0	0.0000	0.0200	0.9377
hsa_steady	kappa_0=0	0.1000	0.2000	1.9083
hsa_steady	delta=0	0.0000	0.0200	1.0322

12.4 Time-Varying Kappa Path



12.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9846	0.6458	-225.9354	1.0000	NaN	NaN	NaN
hsa_dynamic	-120.9323	0.6411	NaN	NaN	NaN	0.3289	NaN
hsa_full	-115.9144	0.6143	NaN	NaN	1.0576	NaN	0.9377
hsa_steady	-121.9385	0.6457	-288.4494	0.0000	1.0322	NaN	NaN

13 inv_markup: prior=weak, period=full

Condition. data=inv_markup, prior=weak, period=full, constraint=unrestricted (unrestricted).

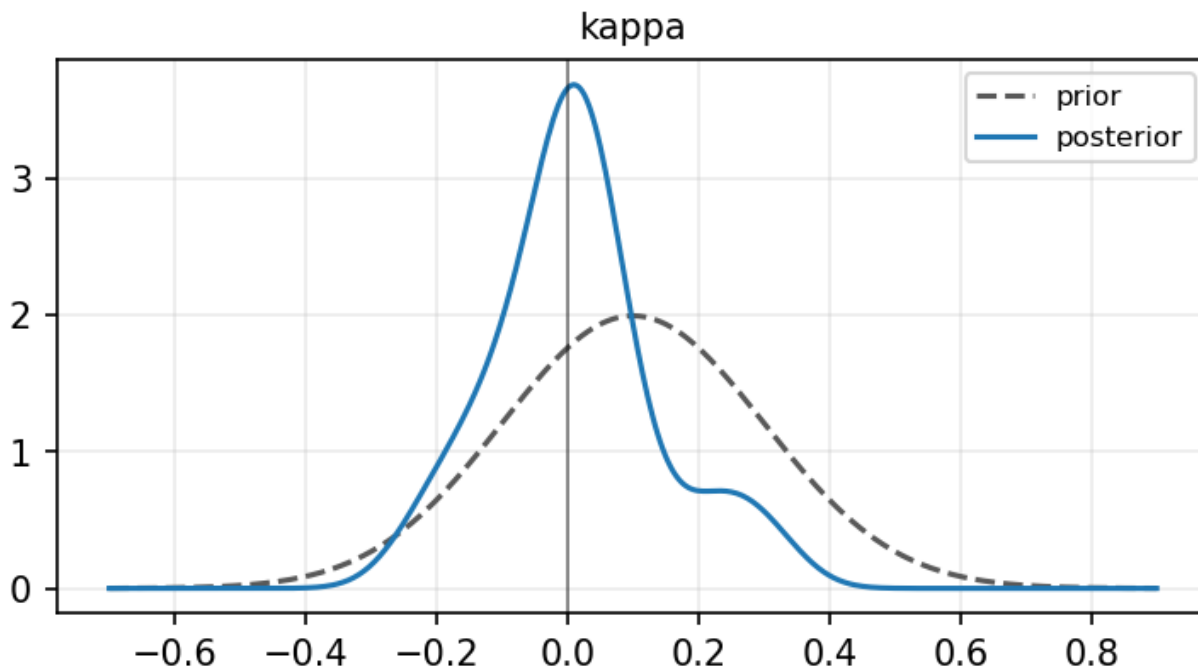
13.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.634 [0.547, 0.743]	0.648 [0.554, 0.776]	0.616 [0.529, 0.724]	0.617 [0.526, 0.720]
kappa	0.012 [-0.205, 0.302]	NaN	0.030 [-0.213, 0.302]	NaN
kappa_0	NaN	0.101 [-0.188, 0.462]	NaN	0.096 [-0.209, 0.405]
delta	NaN	-0.001 [-0.003, 0.002]	NaN	-0.000 [-0.003, 0.001]
theta	NaN	NaN	0.008 [0.003, 0.016]	NaN
theta_0	NaN	NaN	NaN	0.067 [-0.242, 0.371]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.002]
rho_1	NaN	1.805 [1.706, 1.895]	1.871 [1.807, 1.910]	1.191 [1.002, 1.388]
rho_2	NaN	-0.809 [-0.897, -0.709]	-0.875 [-0.915, -0.810]	-0.269 [-0.470, -0.066]
phi_1	0.673 [0.532, 0.774]	0.667 [0.561, 0.778]	0.651 [0.517, 0.783]	0.664 [0.584, 0.784]
lambda_ez	-0.037 [-0.338, 0.225]	0.020 [-0.184, 0.326]	NaN	-0.012 [-0.301, 0.278]
n	NaN	-0.054 [-0.190, 0.050]	-0.035 [-0.114, 0.051]	-0.090 [-0.181, 0.037]

13.2 Prior vs Posterior by Model

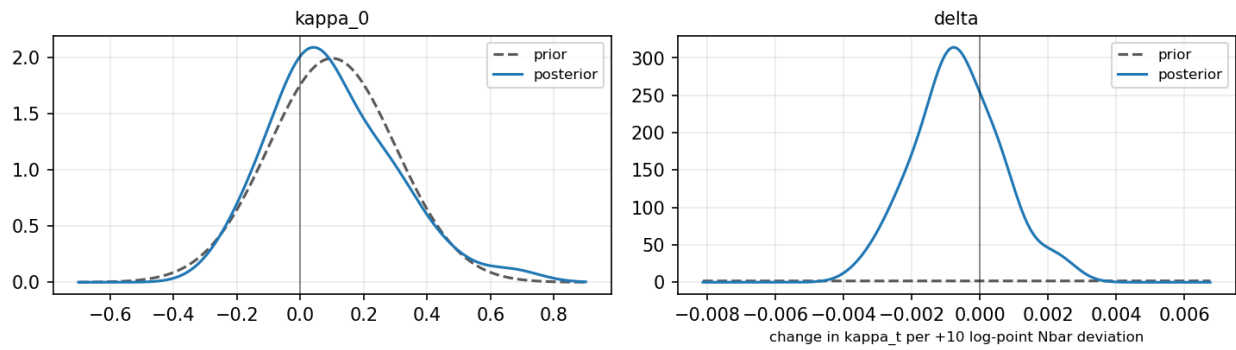
ces

Prior vs posterior — ces



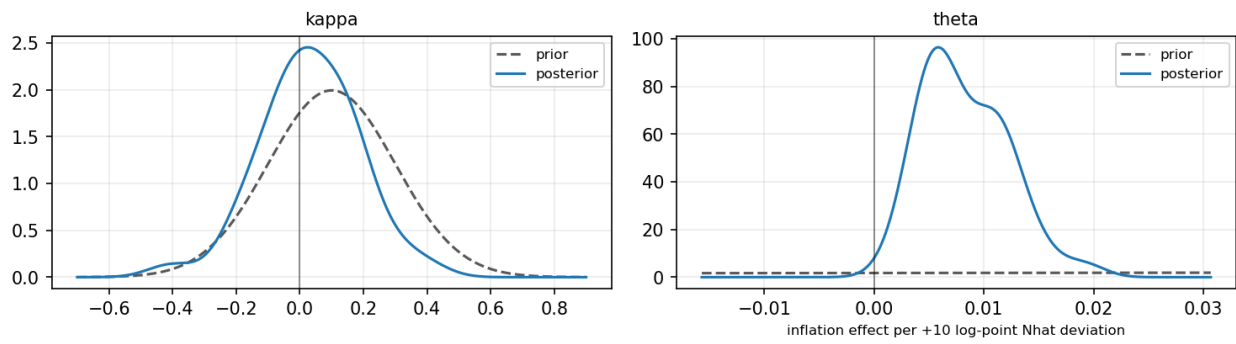
hsa_steady

Prior vs posterior — hsa_steady



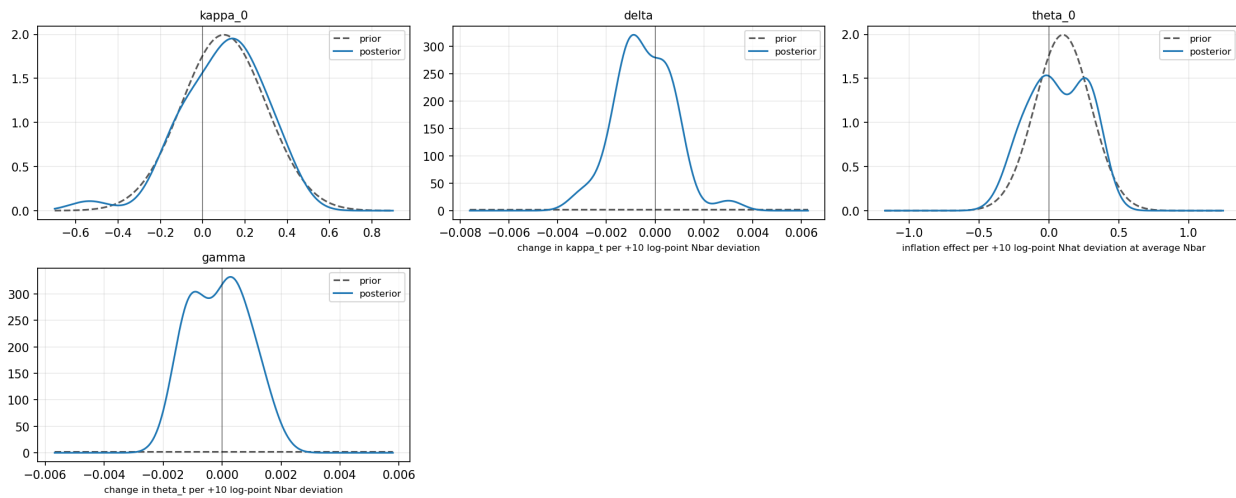
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

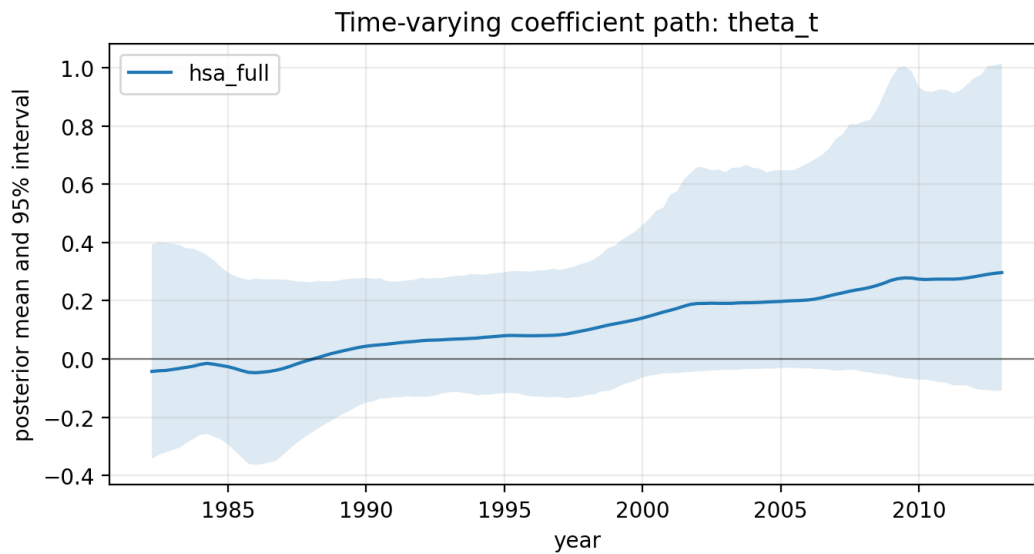
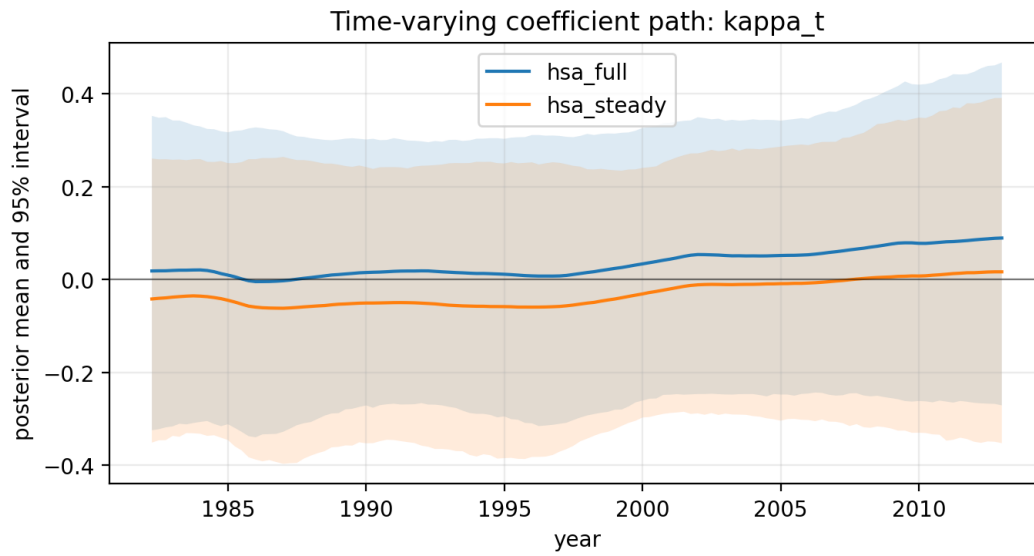
Prior vs posterior — hsa_full



13.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.5000	3.6252
hsa_dynamic	kappa=0	0.1000	0.5000	3.1692
hsa_dynamic	theta=0	0.0000	0.5000	0.2768
hsa_full	kappa_0=0	0.1000	0.5000	3.5665
hsa_full	delta=0	0.0000	0.1000	1.3678
hsa_full	theta_0=0	0.0000	0.5000	2.0650
hsa_full	gamma=0	0.0000	0.1000	0.6427
hsa_steady	kappa_0=0	0.1000	0.5000	3.4524
hsa_steady	delta=0	0.0000	0.1000	1.6061

13.4 Time-Varying Kappa Path



13.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.7977	0.6453	-225.9501	1.0000	NaN	NaN	NaN
hsa_dynamic	-116.6391	0.6196	NaN	NaN	NaN	0.2768	NaN
hsa_full	-119.1551	0.6323	NaN	NaN	1.3678	NaN	0.6427
hsa_steady	-121.6596	0.6445	-280.5963	0.0000	1.6061	NaN	NaN

14 inv_markup: prior=tight, period=full

Condition. data=inv_markup, prior=tight, period=full, constraint=unrestricted (unrestricted).

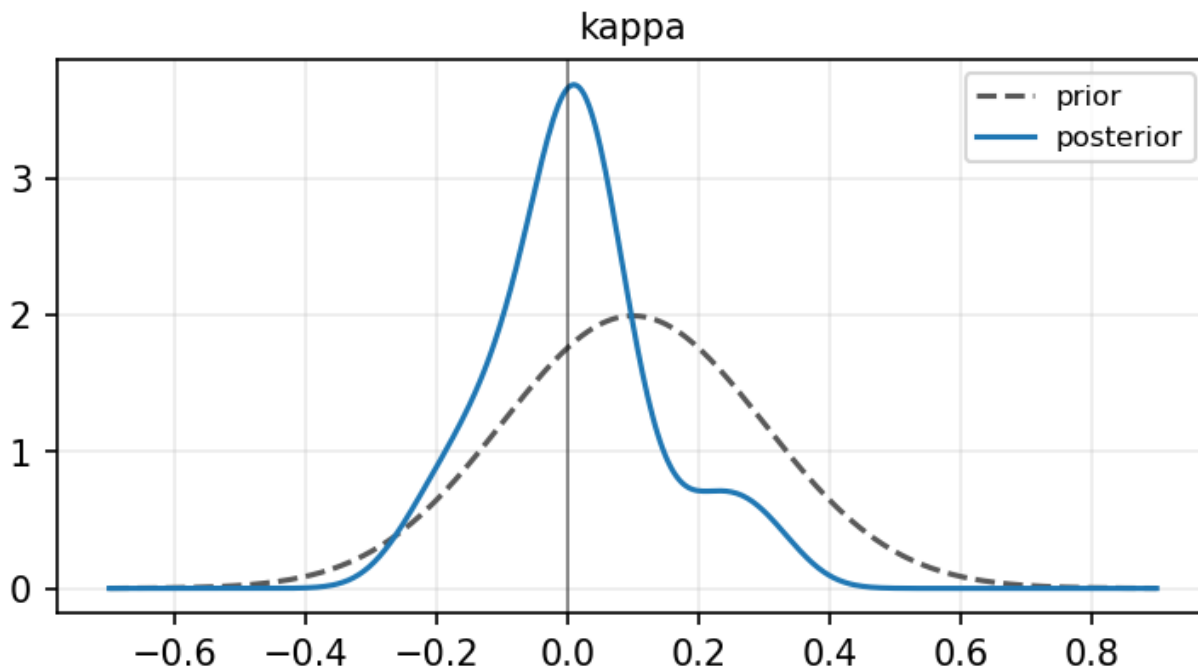
14.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.634 [0.547, 0.743]	0.648 [0.554, 0.776]	0.616 [0.529, 0.724]	0.617 [0.526, 0.720]
kappa	0.012 [-0.205, 0.302]	NaN	0.030 [-0.213, 0.302]	NaN
kappa_0	NaN	0.101 [-0.188, 0.462]	NaN	0.096 [-0.209, 0.405]
delta	NaN	-0.001 [-0.003, 0.002]	NaN	-0.000 [-0.003, 0.001]
theta	NaN	NaN	0.008 [0.003, 0.016]	NaN
theta_0	NaN	NaN	NaN	0.067 [-0.242, 0.371]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.002]
rho_1	NaN	1.805 [1.706, 1.895]	1.871 [1.807, 1.910]	1.191 [1.002, 1.388]
rho_2	NaN	-0.809 [-0.897, -0.709]	-0.875 [-0.915, -0.810]	-0.269 [-0.470, -0.066]
phi_1	0.673 [0.532, 0.774]	0.667 [0.561, 0.778]	0.651 [0.517, 0.783]	0.664 [0.584, 0.784]
lambda_ez	-0.037 [-0.338, 0.225]	0.020 [-0.184, 0.326]	NaN	-0.012 [-0.301, 0.278]
n	NaN	-0.054 [-0.190, 0.050]	-0.035 [-0.114, 0.051]	-0.090 [-0.181, 0.037]

14.2 Prior vs Posterior by Model

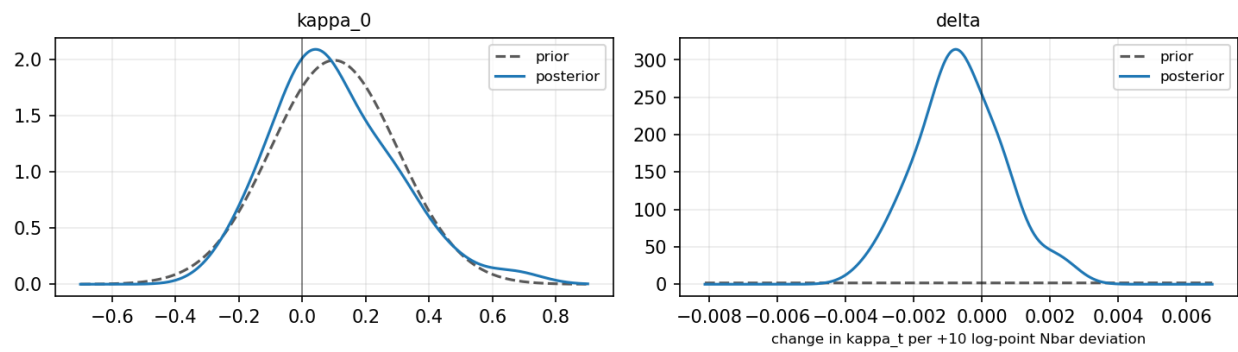
ces

Prior vs posterior — ces



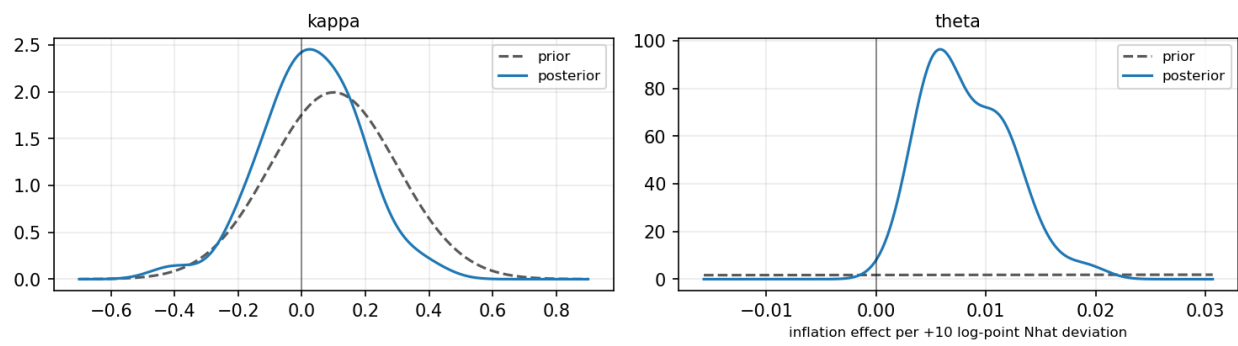
hsa_steady

Prior vs posterior — hsa_steady



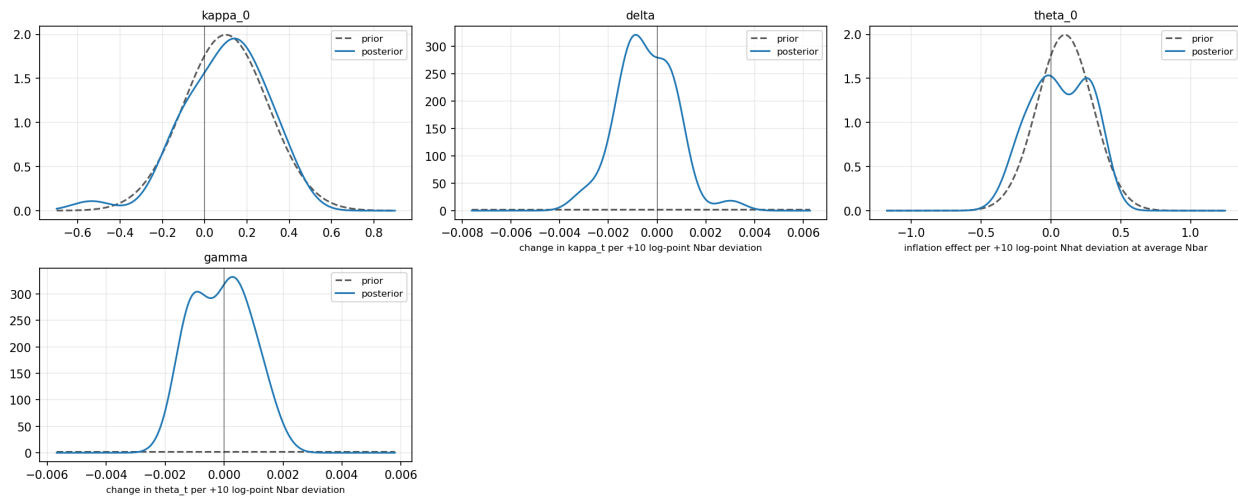
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

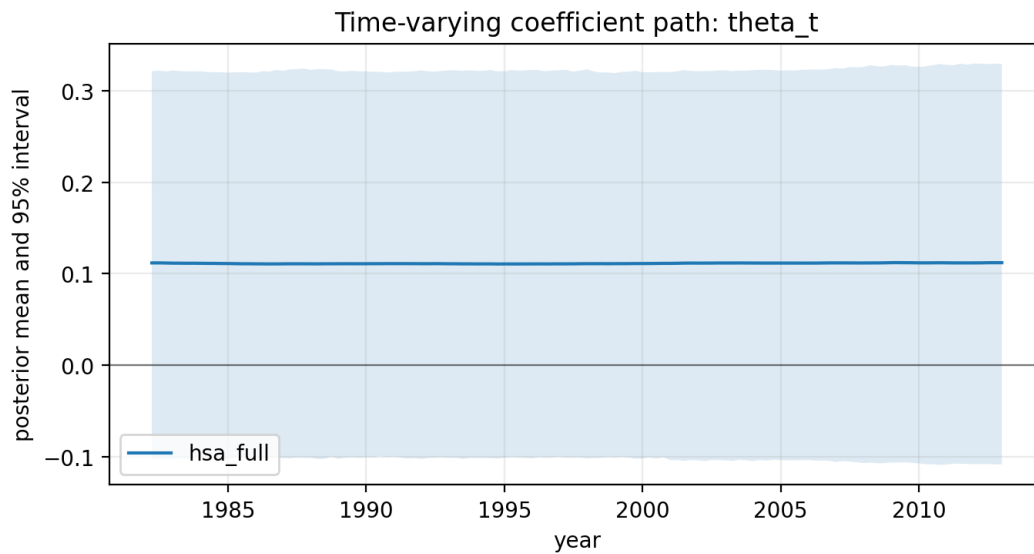
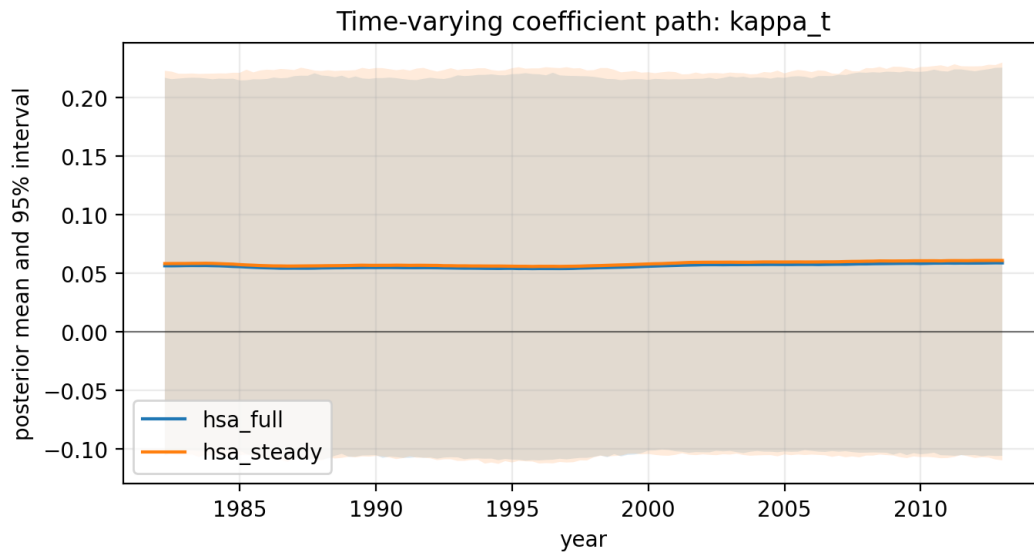
Prior vs posterior — hsa_full



14.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.1000	1.6120
hsa_dynamic	kappa=0	0.1000	0.1000	1.9257
hsa_dynamic	theta=0	0.1000	0.1000	1.0081
hsa_full	kappa_0=0	0.1000	0.1000	1.6374
hsa_full	delta=0	0.0000	0.0100	1.0282
hsa_full	theta_0=0	0.1000	0.1000	0.8813
hsa_full	gamma=0	0.0000	0.0100	1.0296
hsa_steady	kappa_0=0	0.1000	0.1000	1.5858
hsa_steady	delta=0	0.0000	0.0100	0.9580

14.4 Time-Varying Kappa Path



14.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-122.6747	0.6484	-225.9948	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.9861	0.6468	NaN	NaN	NaN	1.0081	NaN
hsa_full	-122.0479	0.6453	NaN	NaN	1.0282	NaN	1.0296
hsa_steady	-122.6619	0.6484	-314.6875	0.0000	0.9580	NaN	NaN

15 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.